

Predicting the Influence of the Padma Bridge on Future Urban Growth

By

Rifat Hossain Talukder

Student ID: 180421



BURP DISSERTATION

Dissertation submitted to Urban and Rural Planning Discipline, Khulna University,
Khulna, in partial fulfillment of the requirements for the degree of Bachelor of
Urban and Rural Planning (BURP)

February 2023

Urban and Rural Planning Discipline
Khulna University
Khulna-9208

Dissertation Acceptance Form

Predicting the Influence of the Padma Bridge on Future Urban Growth

By

Rifat Hossain Talukder

Student ID: 180421

Dr. Md. Ashiq Ur Rahman

Professor

Urban and Rural Planning Discipline

Khulna University, Khulna-9208

**Chairman of the Examination
Committee**

Dr. Md. Ahsanul Kabir

Professor

Urban and Rural Planning Discipline

Khulna University, Khulna-9208

Supervisor



Mostaq Ahmed

Assistant Professor

Urban and Rural Planning Discipline

Khulna University, Khulna-9208

External Evaluator

Acknowledgment

I would like to thank Almighty Allah for his blessings. It is through His grace that I have been able to achieve this milestone and I am forever grateful for His constant presence in my life.

I would like to express my deepest gratitude to my supervisor Dr Ahsanul Kabir, for his patience, guidance, and unwavering faith in me. His guidance and encouragement have been invaluable, and his patience has been greatly appreciated. I am truly grateful for all that he has done for me and for being an integral part of my success with his invaluable contributions. I could not have done it without his dedicated supervision and guidance.

I would also like to express my sincere gratitude to my external supervisor, Mostaq Ahmed, for his soothing advising and encouragement. His insights and expertise have been instrumental in shaping my work and I am truly grateful for his support and guidance throughout my journey

I would also like to extend my heartfelt thanks to my parents for their love and support. Their constant encouragement and belief in me have been a source of strength and motivation throughout my journey.

Rifat Hossain Talukder

February, 2023

Abstract

The newly built Padma bridge is supposed to reshape the economy of the southern region of Bangladesh. Although the economic and structural benefits of the structure are estimated in some studies, but its impact on the physical growth of built-up area is still not assessed. To assess any impact of anything, a study of impact patterns is necessary. But, as a newly built structure, the Padma bridge is yet to put a recognizable pattern. So, in this study, I have assessed another similar structure, Jamuna Bridge. This research studies the growth pattern in the region of Jamuna bridge before and after the construction of the bridge using satellite images and machine learning algorithms. Then it is used as a training model while adjusting the context based on the difference of growth pattern in Padma and Jamuna region before construction. To assess the growth, this study used Markov chain models to evaluate transition potential matrix and Cellular automata model found in Dinamica EGO to predict growth. After adjusting the model with different urban conditions based on the existing pattern and the pattern after the growth, The difference between these two outcomes is seen as the impact of the Padma bridge on urban growth. The study shows, in the next 24-years there is almost 11.87% additional growth expected for the construction of the Padma bridge. Which, in terms of area, is almost 317 sqKM. Gopalganj Sadar and Madaripur Sadar along with Shib Char are expected to experience the highest influence while the Hizla, Muladi, and Char Bhadrasan are supposed to experience least change because of the Bridge's Construction.

Keywords: Urban Growth, Growth Simulation, Padma Bridge, Machine Learning Based Growth Detection

Table of Contents

Chapter 1: Introduction

1.1	Background	1
1.2	Aim	3
1.3	Objectives.....	3

Chapter 2: Literature Review

2.1	Expected Impact of Padma Bridge.....	5
2.1.1	Improved Connectivity	5
2.2	Economic Prospects	5
2.3	Trading.....	6
2.4	Overall Urban Context	6
2.5	Tools for Urbanization Extraction from Remotely sensed data	7
2.6	Pixel Based Classification	8
2.6.1	K-means and ISODATA Cluster	8
2.6.2	Object Based Classification	8
2.6.3	Change Detection.....	9
2.7	Tools for Urbanization Prediction	11
2.7.1	Cellular Automata (CA).....	11
2.7.2	Artificial Neural Networks (ANN)	11
2.7.3	Fractal Modeling.....	11
2.7.4	Agent-based Modeling.....	12
2.7.5	Decision Tree Modeling.....	12

Chapter 3: Materials and Method

3.1	Study Area	17
-----	------------------	----

3.2	Methodology.....	18
3.3	Data Used	20
3.4	Variables for Prediction.....	23
3.5	Selecting the Most Preferable Method	24
3.6	Land Use Classification	27
3.6.1	Training the Model.....	27
3.6.2	Employing Classification Models	29
3.6.3	Extracting Urban Area.....	34
3.7	Predicting Urban Growth.....	36
3.7.1	Prediction model	36
3.7.2	Creating Growth Models	40

Chapter 4: Analysis

4.1	Outcomes from the models.....	43
-----	-------------------------------	----

Chapter 5: Conclusion

5.1	Conclusion	56
-----	------------------	----

References 57

Appendix

Metadata of the satellite image.....	i
R code for Classification	vii

List of Figures

Figure 3.1. Southern Upazilas within a radius of 50km from Padma Bridge	17
Figure 3.2. Methodology diagram	19
Figure 3.3. Images for Padma Bridge Area (Sequenced by date)	22
Figure 3.4. Images for Jamuna bridge Area (Sequenced by Date)	22
Figure 3.5. Band-wise Reflectance Distribution for Each Class.....	27
Figure 3.6. Whisker Diagram of Reflectance distribution across Bands	28
Figure 3.7. Output of different Methods	29
Figure 3.8. Overall accuracy of the applied models	30
Figure 3.9. Urban condition of the Jamuna region before Jamuna Bridge	34
Figure 3.10. Urban condition of the Jamuna region after Jamuna Bridge.....	35
Figure 3.11. Growth prediction model flow chart.....	37
Figure 4.1. Comparison of real scenario and Model 1 simulation.	43
Figure 4.2. Similarity graph based on window size for model 1	44
Figure 4.3. Comparison of real scenario and model 2.....	44
Figure 4.4. Similarity graph based on window size for model 2	45
Figure 4.5. Comparison of real scenario and Model 3.....	46
Figure 4.6. Similarity graph based on window size for model 3	47
Figure 4.7. Comparison of real scenario and Model 4.....	47
Figure 4.8. Similarity graph based on window size for model 4	48
Figure 4.9. Similarity graph based on window size for model 5	49
Figure 4.10. Comparison of real scenario and Model 5	49
Figure 4.11. Comparison of real scenario and Model 6	50
Figure 4.12. Similarity graph based on window size for model 6	50
Figure 4.13. Comparison of real scenario and Model 7	51
Figure 4.14. Similarity graph based on window size for model 7	52
Figure 4.15. Predicted Growth of 2046 and Transition probability	53

Figure 4.16. Predicted impact growth of Padma bridge	54
Figure 4.17. Upazila-wise Padma Bridge influenced Predicted growth within 2046.	55

List of Tables

Table 3.1. List of used satellite images	21
Table 3.2. Variables used for prediction	24
Table 3.3. Advantages and Disadvantages of different methods	25
Table 3.4. Skewness of the bands' distribution	28
Table 3.5. Accuracy of the classifications	33
Table 3.6. Urban changes in different timeframe.....	35

Chapter 1: Introduction

1.1 Background

As a riverine country, Bangladesh has an enormous number of (approx. 700) large and small rivers. The Padma, Brahmaputra-Jamuna, and Meghna rivers divide the nation into four primary zones: North, East, Northwest, and Southwest. In addition to the Jamuna Bridge, there are several significant bridges connecting the Northwest, North, and East zones to the capital Dhaka. However, the Padma River still separates the Southwest Zone from important destinations like Khulna, Mongla, and the main land port at Benapole. (Sham & Tapley, n.d.)

In order to foster economic, social, and industrial development in the comparatively undeveloped southwestern zone, the Padma Bridge across the Padma River is intended to physically link the nation. People from Bangladesh's southern area will be the most benefitted ones from the Padma Bridge's construction. One may expect a multiplier impact from such infrastructure. The bridge is expected to boost the national economy by 2% while also boosting the south west's by 35%. By creating jobs for 1.2 percent of the entire work force, the bridge is intended to reduce public hardship (Zaman et al., 2006). Its completion will transform the economy, boosting Bangladesh's rating from 100th to more competitive levels in the World Bank's 2018 Logistics Performance Ranking. The Padma multifunctional bridge is expected to reshape our country's environment in ways we cannot fathom (Raihan & Khondker, n.d.).

With the bridge, Supply networks that are robust will allow farmers to retain product quality while lowering production expenses due to transportation and storage. Exporting agriproducts will give quality assurance as export zones become increasingly close. SMEs like small-scale garment and export manufacturing enterprises will grow when transportation and energy limitations are eased. Small and medium-sized businesses (SMBs) will be able to benefit from increasing investment and industrial expansion. Reduced migration will result in less megacity

urbanization and economic decentralization. Which will greatly increase the urbanization in the southwest zone.

While the accessibility is increased by the bridge, people will seek to live in a place where they can maximize their individual interests. Sometimes proletariats are willing to compromise service necessities to maximize their economic vehemence. Oftentimes people choose to reside in the buffer of a new infrastructure to reduce economic expanse of living with a cost of reach to service facilities, and the city grows.

The problem about this growth is, it is unplanned and in future it will be hard to provide quality standard for these areas in many aspects, as they were not prepared for the facilities they will be asking. Exurban low-density areas burn-through more energy per capita than their high-density counterpart nearer to the city's center. Energy for warming, cooking, cooling, lighting, and transportation is generally delivered by consuming petroleum derivatives, a cycle that adds to air contamination and an unnatural weather change. To arrive at their positions in the city or other business territories, numerous suburb laborers should drive via car. Outings to supermarkets or other retail foundations in city center should likewise be finished via automobiles. Air contamination delivered by gas fueled cars can consolidate with different pollutants to frame photochemical exhaust cloud. Quite possibly one of the most overlooked natural impacts of unplanned structure development is the annihilation of untamed living environment. To clear a path for human residences and their related framework, natural land is furrowed under, reviewed, and cleared. Sluggish streams are regularly diverted to give more productive waste to lodging plots and business zones. Though, little zones of wild life natural surroundings may remain, they might be too little to even consider supporting all the local species that lived there previously or might be generally isolated from each other. This course of action frequently powers natural life to cross perilous human-overwhelmed scenes to discover food or mates. Sometimes this growth also affects the existing city

dwellers as the services are shared and decreases the transport efficiency. Moreover, all these settlements often become a problem for future development and planning. And more than often authorities cannot take steps against these because the area is out of city boundary and they cannot know where will the growth grow. Predicting the possible urbanization may help decision makers to plan cityscape.

1.2 Aim

This research aims to analyze the pattern of the growth causing by the Padma bridge and predict the future urban growth of the impacted region.

1.3 Objectives

The overall objective of the research is to explore the land use change on the southern part of the Padma Bridge and to envisage the future land use and its intensity in the years to come. The specific objectives are:

- To assess the impact of Padma bridge on urbanization on the southern side of the bridge
- To predict the future urbanization and land use change influenced by the Padma bridge

Chapter 2: Literature Review

2.1 Expected Impact of Padma Bridge

2.1.1 Improved Connectivity

The Padma Multipurpose Bridge will tackle the communication problem of Bangladesh's southwestern region, which accounts about 25% of the overall land mass. The introduction of both the road and the rail will aid in the transportation of raw materials from the port of Chittagong at a lower cost.

By completing the bridge, the southwest's economic growth will stimulate industrial and commercial activities while also improving economic and job possibilities for area residents. The planned bridge will give direct linkages to the country's two major seaports. The distance from Dhaka to practically all important locations in the southwest area will be lowered by 100 kilometers (km) or more, and vehicle hauling time will be cut by more than 3 hours each trip, resulting in significant savings in passenger and commodities transportation time and expenses. (Islam, 2020)

2.2 Economic Prospects

The economic output of the transportation, commerce, and regional industrial revolution focused around the Padma Bridge, as well as GDP growth, will have an influence on the country's entire economy. According to a Dhaka Tribune article, the Padma Multipurpose Bridge's opening is estimated to boost the country's GDP by 1.3 to 2% every year. At the same time, the completion of the Padma Bridge Rail Link project will increase GDP by 1%. A BRAC research anticipates a 5% contribution to GDP in 31 years based on the present base year. However, once completed, the bridge will add 1.2 percent to the country's GDP. (Rahman & Khondker, 2022)

2.3 Trading

The inauguration of the Padma Bridge is expected to increase trade across the country. Experts predict an industrial revolution in 21 southwest districts, particularly within a year after the opening of the double-decker bridge. A freight truck takes at least two to three days to go from Barisal or Khulna to Dhaka or from Dhaka to these places. After the bridge opens, this period will be shortened to a maximum of one day, increasing regional trade by many times. According to a Daily Star source, after the Padma Bridge Rail Link project is done, the travel between Dhaka and Kolkata via Jessore would be cut in half, taking only 3 to 4 hours, boosting the country's foreign trade. As a result, Bangladesh may become an important international trade corridor. According to research by the Khulna Chamber of Commerce, after the Padma Bridge is completed, the region would play a pioneering role in South Asian regional connectivity, connecting Dhaka to the ports of Mongla and Payra. Now, the Chittagong Port handles more than 90% of the country's foreign commerce. This port's commerce volume was around 90 billion dollars in 2021. Initially, Chittagong's port could accommodate 2 million TEU (Twenty-Foot Equivalent Unit) containers per year, but by 2021, that figure had risen to more over 3.2 million TEU. (Rahman & Khondker, 2022)

2.4 Overall Urban Context

The southwestern region's development has been hampered by the lack of a bridge. Because of the transportation and result-marketing problems of both raw materials and completed goods, no industry could be formed save those relying on local resources. Now that the Padma Multipurpose Bridge will deliver the electricity required for industrialization to the doorsteps of ordinary people,

industries of all scales and dimensions will flourish in the region, ultimately strengthening the national economy.

Natural gas and notional power grid will be connected to the southwestern region through the bridge. The two parts will be connected via an electric circuit. As a result, this massive power supply will fuel both the current economy and rapid modernization. (Islam, 2020)

2.5 Tools for Urbanization Extraction from Remotely sensed data

Extracting urbanization is a process of land classification from remotely sensed data. Currently several objects based and pixel-based models are being used for image classification. Unlike pixel-based classification approaches that categorize individual pixels directly, object-based classification combines image pixels into spectrally homogeneous image objects using an image segmentation algorithm before classifying the individual objects. The distinctions between object-based and pixel-based classification approaches may be seen through two lenses: classification units and classification features. On the one hand, the object-based method provides benefits in these two areas over the pixel-based approach. For starters, switching from pixels to picture objects minimizes within-class spectral variance and, in general, eliminates the salt-and-pepper effects that are common in pixel-based classification. Second, a broad number of characteristics defining objects' spatial, textural, and contextual aspects may be obtained as supplemental information to direct spectral observations, possibly improving classification accuracy. (Guo et al., 2007)

2.6 Pixel Based Classification

2.6.1 K-means and ISODATA Cluster

Abbas et al explored K-means and ISODATA to analyze the presentation of unsupervised classification algorithms. The study used NDVI, green, NIR and FIR to classify land cover data into five different classes. i.e., waterbody, agro-based, settled, forest and barren land. For the K-means cluster, the target is to reduce the summation of squares in term of errors, between each pixel and the representative cluster is minimized, to measure the summation of squared distances,

$$SS_{distances} = \sum_{Ax} [X - C(x)]^2$$

And for Mean Squared Error, $MSE = \frac{SS_{distance}}{(N-c)^b}$ formula is used,

Where, $C(x)$ indicates the mean of the cluster at x pixel, N is the number of pixel and c indicates the number of clusters.

Where in ISODATA is a consistent adjustment of class means rather than band pixels using least distance approaches. In the K means approach the number of K remains the same throughout the iteration, although it may turn out that fewer cluster could fit the data better. Which is not the problem in ISODATA algorithm. It allows the number of clusters to be adjusted automatically during the iteration as it merges and splits clusters with higher standard deviations.

Abbas et al concluded that, the unsupervised classification of land patterns with each algorithm varies upon the number of iterations and number of classes. Different degrees of accuracy was yielded and all were justifiable by their causes. (Abbas et al., n.d.)

2.6.2 Object Based Classification

Li et al explored a variety of algorithms to compare them based on their accuracy. All the algorithms ran against same classification task to ensure the fairness of the

comparison. Landsat Thematic mapper dataset was used over Guangzhou City of China. Among the 13 unsupervised classifications algorithm, their vision was primarily on the spectral information provided by TM data. They found that, when sufficiently representative training samples were used, most of the algorithms performed reasonably well. Among the algorithms they used, Stochastic Gradient bound, Random Forest and Decision tree (C4.5) performed slightly better than other models. (Li et al., 2014)

2.6.3 Change Detection

Jensen explored various ways of photointerpretation. He found that digital change detection particularly focuses on the method wise collection of data from a large area. Several methods like image differencing, image rationing, support vector machine etc. was discussed in his study. He concluded that it is highly tough to maintain precision on digital change detection. But as the digital detection is less exhaustive and easier to replicate it is more viable to use these methods.

In his comparison, post classification failed to omit where it should and detected too much changes. Where clustering comparison misinterpreted between classifications. (Jensen, 1981)

Lu et al., did a similar analysis among different change detection methods. Where he found, in Image subtraction method, The first date image is subtracted from the second date image, pixel by pixel. It is an easily interpretable straightforward method but a detailed change matrix cannot be provided while this method is used.

For regression function, it calculates the intensity values of given second-date image and subtracts them from the first-date image, establishing a link between the two images. He found, change detection involves developing correct

regression functions for specified bands before implementation but also is inaccurate in many circumstances despite its ability to lessen the influence of meteorological, sensor, and environmental variations between two dates.

For Image ratio, it determines, band by band, the ratio of two dates of the recorded images. Reduces the negative effects of topography, shadow, and sun angle. Because of this, the distribution of results is frequently questioned.

About PCA model, he stated, change information may be found in the new components of multi-temporal data derived from PCA. PCA can be used to detect changes in images in two ways: (1) by combining images from different dates into a single file, and (2) by performing PCA separately and subtracting the resulting PC image from the second date from PC image from the first date in order to extract change information from the principal component images. In the derived components, less data is duplicated across bands, allowing for more distinct information to be shown. PCA, on the other hand, is scene dependent, making it challenging to understand and classify the findings of change detection across various dates. A full matrix of change class information is not available and needs defining criteria to identify the change areas.

For the Artificial Neural Networks, the spectral data of the period of change is utilized to train the neural network. The multi-layer perceptron neural network model is often trained using a backpropagation algorithm. Using the training samples, it is possible to estimate the attributes of data using ANN. The problem is that the nature of the hidden layers is unknown and requires a significant period of training to master. The quantity of training data used in an ANN model may have a significant impact on the accuracy of the model.(Lu et al., 2004)

2.7 Tools for Urbanization Prediction

2.7.1 Cellular Automata (CA)

Batty and Longley described the rudiments of Cellular automata, where the stated, CA are characterized as discrete dynamical systems with cells on a grid, in which local linked linkages display global changes A cell's current state is often determined by its previous state's value as well as the values of the cells around it, according to a set of transition rules. Depending on how supportive or limiting environmental and socioeconomic factors are, these regulations have an impact on urban expansion. By simulating local activities and observing the resulting global emergent structure, the bottom-up strategy used in CA can achieve its goals because of the repeated nature of the CA process and the resulting fractal designs (Batty, 2005)

2.7.2 Artificial Neural Networks (ANN)

The ANN learns from the data it already has about input and output. It does this in an "iterative" way (i.e., back-propagation algorithm). ANNs have become more popular in recent years because of better computer power that can be used in a wide range of scientific fields(Aisa et al., 2008). It is because cities grow in a big way that the ANNs learning process can make tools that can model the complexity of urban structures. (Wu et al., 2006)

2.7.3 Fractal Modeling

Fractal geometry has also been used to predict how cities will grow. Classical Euclidean geometry is not good enough to describe how things move in space and time in nature. (Mandelbrot & Mandelbrot, 1982) came up with fractal geometry, and since then, fractals have been used in many different scientific fields. Cities can be thought of as fractal objects, where the interaction of different parts of the space can be described in non-linear ways (Batty, 2005). Fractal theory talks about

the non-linearity of spatial structural complexity, which shows that cities grow in a way that is multiscale and self-organizing (Shen, 2002). Self-organization is a big part of what happens in the world. It is based on how well the system can put together its parts with help from its own power source. In self-organized systems, the organization grows on its own without being controlled by an outside force. It is caused by internal changes, which could be fluctuations or noise. When some parts of the city are isolated and looked at separately, the urban environment behaves in an unusual way. Because every region is part of the urban self-organized system, it must be treated rather than as a separate part

2.7.4 Agent-based Modeling

Agent-based models use a bottom-up strategy to explain urban systems by simulating individual agent behaviors and quantifying the subsequent system behavior (Crooks, 2006). Agents are self-contained entities that communicate with other agents to exchange data. Individual agent behavior enables the model to integrate the effect of human decision making.

2.7.5 Decision Tree Modeling

Decision trees are a top-down classification technique that has been used to simulate land use change (Chan et al., 2001) and to classify land uses using remotely sensed images. Despite their limited application in urban development modeling, decision trees are of interest owing to their capacity to produce rules and the ease with which the model structure may be understood. Decision trees provide a hierarchy of partition rules automatically, which are used to divide data into consecutive pieces. Three fundamental stages are required to develop a decision tree. The first stage is concerned with tree construction through recursive node splitting. The second phase involves a pruning procedure that results in the production of smaller trees with less complexity (Kim & Koehler, 1994). Pruning may also be used to reduce overfitting by removing noisy and

erroneous data. Finally, the tree with the lowest testing error is chosen as the optimum tree. According on the learning algorithm, decision trees are classified into two types: classification trees and regression trees. In the former, the predicted variable's output takes just two values, typically 0 and 1, but in the later, the expected output fluctuates between the dependent variable's values (Osei-Bryson, 2007). Decision trees are commonly employed in urban land use image-based categorization to map urban buildings and plant cover, both of which are critical components of urban modeling and planning (Schneider et al., 2003).

Chapter 3: Materials and Method

3.1 Study Area

Padma bridge is the largest bridge built in Bangladesh till date. And the economic, social, and industrial opportunity it offers is supposed to be life changing for the southwestern part of Bangladesh.. Upazilas within the 50km radius from the Padma multipurpose bridge on the southern side are selected for this research. The study area is bounded by Faridpur Sadar on north, Gopalganj Sadar in the west, Muladi on south and the Padma River itself on the east.

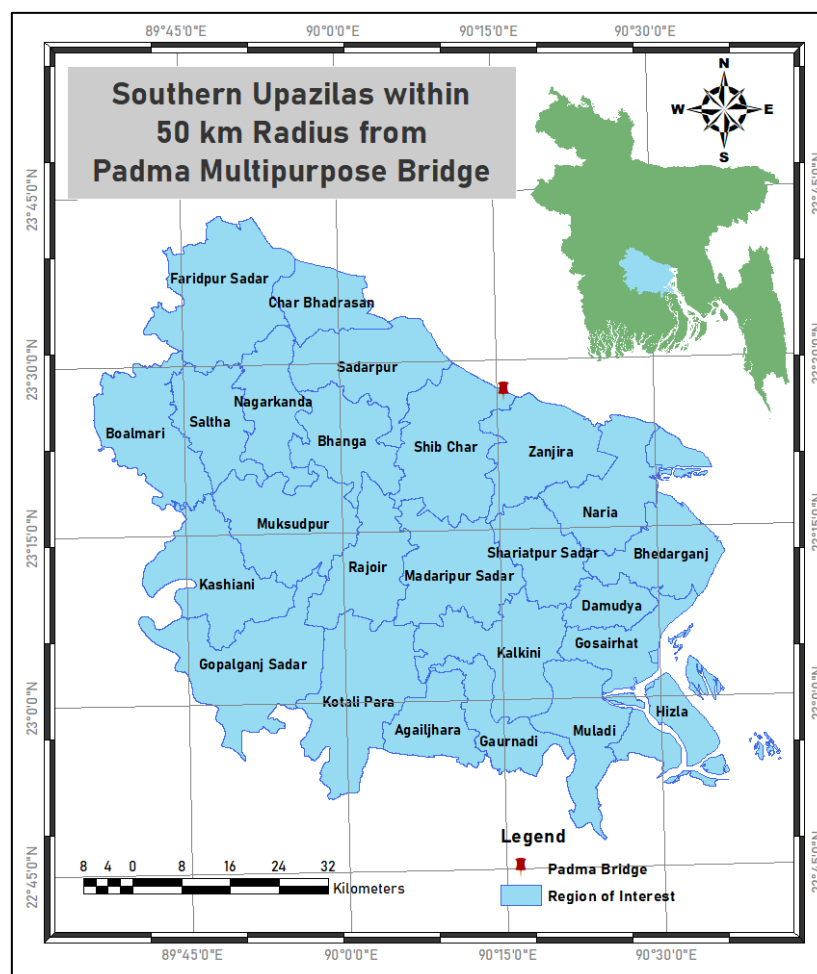


Figure 3.1. Southern Upazilas within a radius of 50km from Padma Bridge

3.2 Methodology

The research is designed considering, the impact of the Padma bridge is objectively distinguishable by predictive modeling. The general idea is, if the physical context forecasted for a specific time in the future using the pre-Padma-bridge period's growth pattern and then again with the post Padma-bridge period's pattern, the difference will be the impact of Padma bridge.

Satellite images from both regions of different timespans are collected which can be separated into 3 classes, pre-Jamuna bridge time, post-Jamuna bridge time, and pre-Padma bridge time. Training points for each dataset were selected using ArcGIS software. The training datasets were then split into two parts for training and for testing. The images are then classified using different machine learning algorithms to separate built-up and non-built-up areas. The algorithm that provides best outcomes are then selected for further analysis.

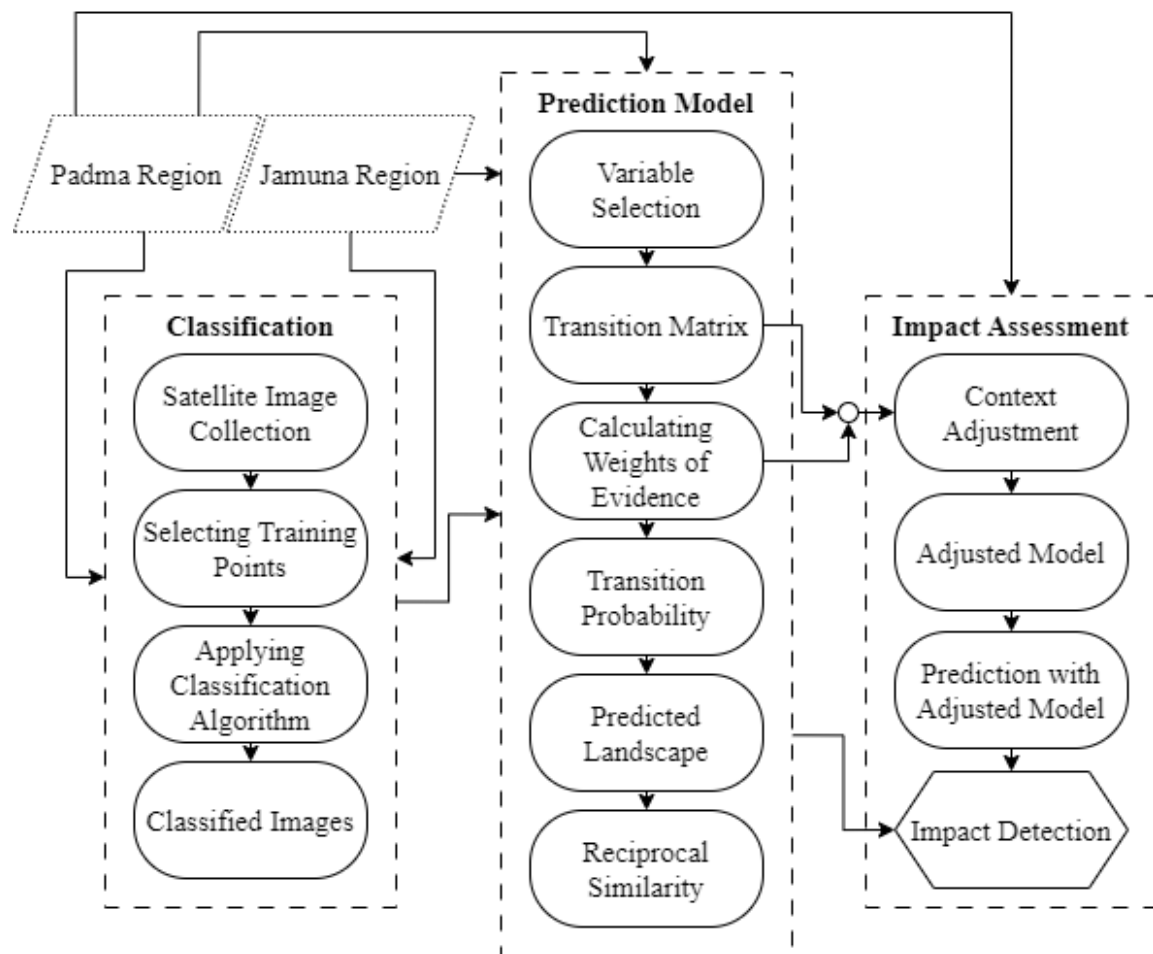


Figure 3.2. Methodology diagram

For prediction model 13 variables were used (Table 3.2). For the growth rate modeling, Markov chain model was used as the urban growth is almost never linear and the change of form depends on its current state not the state in 6 years back. Markov chain model helps to model growth for individual unit of time and for the prediction of next step, the current prediction step is used as initial state. Using weights of evidence method what is the weight of each variable for each distance category is calculated. Combining all the variable's weights, provides the probabilistic map. Where the coefficients are high in average, the growth probability is high there. Using the probability and transition matrix, Expander and patcher functions are used using the agent-based model, Cellular Automata. The cell-based prediction is then formed. To, assess the accuracy, Reciprocal

similarity is used. It compares each state with each other and increases the cell size to search for similar counts. Suppose, 50% accuracy in 5x5 window will mean, the 50% of the predicted cells fall within a 5x5 window radius of the corresponding cell. As, accurately predicting growth at fine scale like 30m is virtually impossible, this method is widely used to check growth prediction accuracy (Pontius Jr & Schneider, 2001).

When pre-Jamuna bridge and post-Jamuna bridge outcomes are compared, the change in transition and weight matrices were noticeable. The transition matrices and weights of evidences are adjusted using geometric difference adjustment method. Using similar method, context was also adjusted for Jamuna to Padma and for pre-Padma to post-Padma contexts. The difference between the prediction based on pre-Padma or Business as usual model with the adjusted model is considered as the impact of The Padma bridge.

The research is not ignorant about the non-spatial or subjective causes inducing urban growth, rather it assumes those variables will either be constant; which will not impair the analysis as it will be like pre-Padma bridge model or those variables are changed due to the construction of the Padma Bridge. In which case that will be the passive impact of Padma bridge. The concern is rather whether if the Padma bridge will have similar impact like Jamuna bridge. In that case, the difference of impact is yet to be known and the research essentially tries to comprehend, what will be the condition if the Padma had a similar impact like the Jamuna bridge.

3.3 Data Used

Longitudinal satellite images were collected from the United States Geological Survey database. Images from different versions of Landsat satellites were taken of different timespan. For image collection, images with minimum cloud cover were chosen and most of the images have 0% cloud coverage. Images with path

no. 138 and row no. 043 represents the surrounding of Jamuna multipurpose bridge and images with path no. 137 and row no. 044 represents the interested region of Padma multipurpose bridge. For the Jamuna bridge; best quality image within a range of approximately 6 years is taken and for the Padma bridge the range is 4 years.

Table 3.1. List of used satellite images

Date of Acquisition	Satellite	Spectral Resolution	Spatial Resolution (m)	Image Quality	Cloud Cover (%)	Path/Row
2010-01-30	Landsat 5	7 Bands	30, 120 (Thermal)	9	0.00%	137/044
2014-03-30	Landsat 8	11 Bands	30, 100 (Thermal)	9	0.00%	137/044
2018-02-21	Landsat 8	11 Bands	30, 100 (Thermal)	9	0.05%	137/044
2022-03-19	Landsat 8	11 Bands	30, 100 (Thermal)	9	0.01%	137/044
1992-03-08	Landsat 5	7 Bands	30, 120 (Thermal)	7	0.00%	138/043
1998-03-09	Landsat 5	7 Bands	30, 120 (Thermal)	9	0.00%	138/043
2003-12-04	Landsat 5	7 Bands	30, 120 (Thermal)	7	0.00%	138/043
2009-03-07	Landsat 5	7 Bands	30, 120 (Thermal)	9	0.00%	138/043
2016-01-06	Landsat 8	11 Bands	30, 100 (Thermal)	9	0.00%	138/043
2022-01-30	Landsat 9	11 Bands	30, 100 (Thermal)	9	0.00%	138/043

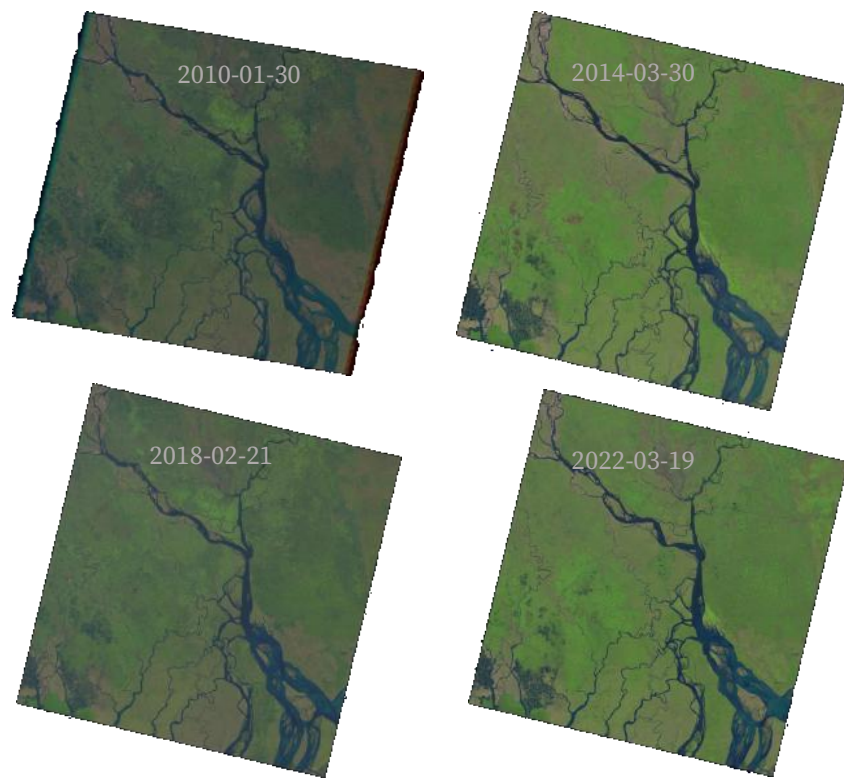


Figure 3.3. Images for Padma Bridge Area (Sequenced by date)

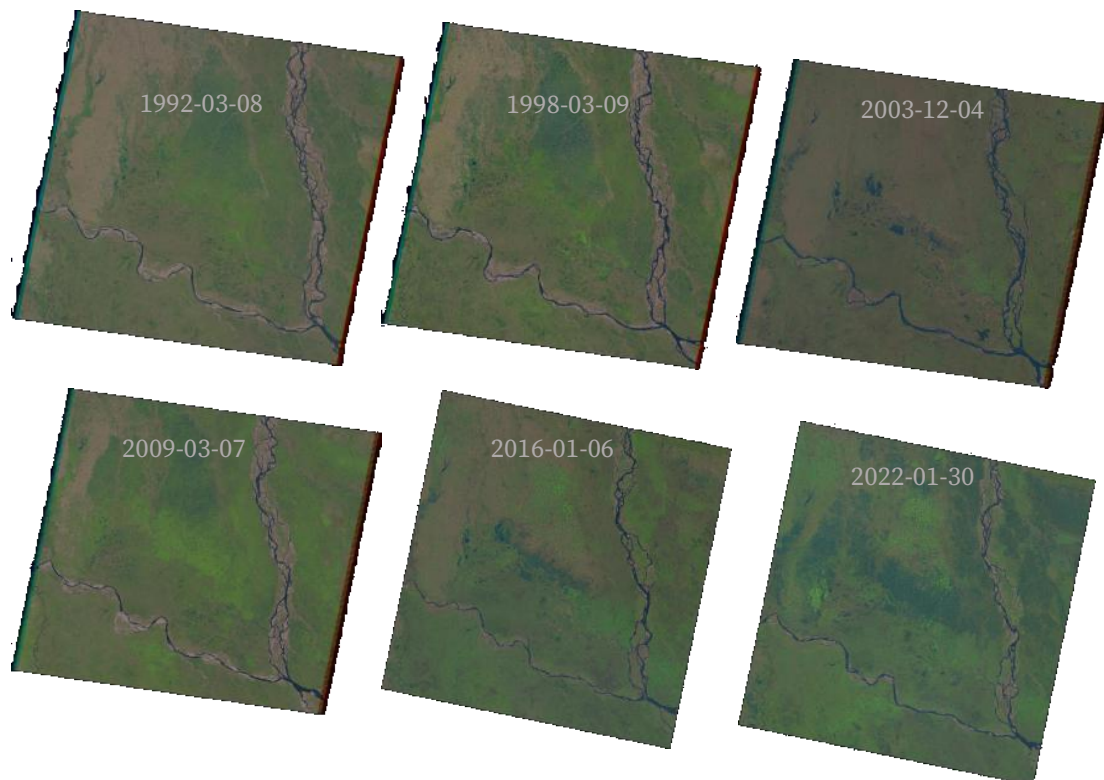


Figure 3.4. Images for Jamuna bridge Area (Sequenced by Date)

3.4 Variables for Prediction

The variables used for the prediction models were spatial variables mostly as distance-based variables. For the distance from roads data, two sources were used. One is derived from Local Government Engineering Department (LGED) database (updated on September, 2018) and other one is Open Street Map (OSM) public database (update of November, 2022) as the former one is not most recent and lacks recent features, it was used for prediction models of earlier dates. OSM data is only used for models of latter dates where there is a feature available in OSM data but not in LGED data, and both were merged afterwards. Road category is defined on the LGED dataset. For nodes, Node's category is derived from the features where it intersects with the features of its own level or lower level. For Level 1 Node, it is defined by, where National roads meet another national road or any regional, zilla or upazila roads. Same goes for the other types of nodes. For Level 5 Nodes, it is defined where Union/Village roads intersected one of its own kind.

Table 3.2. Variables used for prediction

Variable	Data Type	Source
Distance from National Roads	Continuous	LGED (2018), OSM (2022)
Distance from Regional Roads	Continuous	LGED (2018), OSM (2022)
Distance from Zila Roads	Continuous	LGED (2018), OSM (2022)
Distance from Upazila Roads	Continuous	LGED (2018)
Distance from Union & Village Roads	Continuous	LGED (2018)
Distance from Level 1 Nodes	Continuous	LGED (2018)
Distance from Level 2 Nodes	Continuous	LGED (2018)
Distance from Level 3 Nodes	Continuous	LGED (2018)
Distance from Level 4 Nodes	Continuous	LGED (2018)
Distance from Level 5 Nodes	Continuous	LGED (2018)
Elevation	Continuous	SRTM (2000,2013), ALOS (2021)
Hillshade	Continuous	SRTM (2000,2013), ALOS (2021)
Excluded/ Constrain	Dichotomous	Author (2022)

3.5 Selecting the Most Preferable Method

To select the most preferable method for the study, pros and cons of the popular methods are compared against each other.

Table 3.3. Advantages and Disadvantages of different methods

Method	Advantages	Disadvantages
K-Nearest Neighbors (KNN)	i. Easily implementable ii. Can handle multimodal classes iii. Remembers all the training data iv. Can learn instantly v. New data can be added seamlessly (Kamusoko, 2019; Kuhn & Johnson, 2013)	i. Computing the distance for k-nearest neighbors is required ii. Computationally intensive for large data iii. Normalization of data is necessary iv. Sensitive to missing values or outliers (Kamusoko, 2019; Kuhn & Johnson, 2013)
Artificial Neural Network (ANN)	i. Normal distribution is not necessary ii. Can generate non-linear decision boundary iii. Can learn complex patterns iv. Training model errors does not compromise output significantly v. Can evaluate the learned target very fast (Berberoglu et al., 2000; Tso & Mather, 2009)	i. Susceptible to overfitting ii. Difficult to design effective network iii. Lacks interpretability, gives result but cannot explain why and how. iv. ANN works with numeric values. Non numeric data can lose integrity in translation. (Kavzoglu & Mather, 2003; Tso & Mather, 2009)
Single Decision Tree (DT)	i. Both categorical and numeric data are easily integrated ii. Less training required iii. Normal distribution not necessary (Miao et al., 2012; Sesnie et al., 2008; Tso & Mather, 2009)	i. Requires large training samples. ii. Susceptible to small changes in training data. iii. Outliers reduce model reliability (Kamusoko, 2019; Sesnie et al., 2008)
Support Vector Machine (SVM)	i. Is not bound to the normal distribution assumption ii. Performs well when distinguishable	i. Is not proven efficient for large data sets ii. Susceptible to noise in training data

	boundary between class is set. iii. In high dimensional spaces, it scales relatively well iv. Complex problems can be solved using an appropriate kernel function. (Hsu et al., 2003; Kuhn & Johnson, 2013)	iii. Large training data is required iv. It cannot provide any probabilistic explanation for the classification (Hsu et al., 2003; Zhang & Ma, 2008)
Random Forest (RF)	i. Large number of numeric or categorical data can be handled simultaneously ii. Not bound to normal distribution assumption iii. Can generate robust outcome against noise and outliers iv. Can handle missing data effectively v. Overfitting issue is addressed vi. Does not require a hyper-parameter tuning for a reasonable prediction (Breiman, 2001; Rodriguez-Galiano et al., 2012)	i. Interpreting this model is somewhat complex. ii. Biased in favor of variable with different level of categories. iii. Comparatively more computational resource is required (Kamusoko, 2019; Tso & Mather, 2009)

After analyzing the pros and cons, Clearly, Random Forest seems to be the most forgiving model for training error and previous literatures also showed that this model provides adequate accuracy. However, all the models were applied with the same training dataset to check which one performs the best.

3.6 Land Use Classification

3.6.1 Training the Model

Though the random forest algorithm is forgiving to atmospheric interference and lousy training data, still a good number of training sample is needed to perform a comparatively accurate data. To train the classification model, training area from each date is needed to be selected for each class that are spatially scattered and covers the overall image. Roughly 1000 samples for each class are selected for this study. The training data is then divided into 2 parts, where 85% will be used for training the algorithm and other 15% will be used to assess the accuracy of the model outcome.

Distribution of bands for different classes in the 2022 image are shown below:

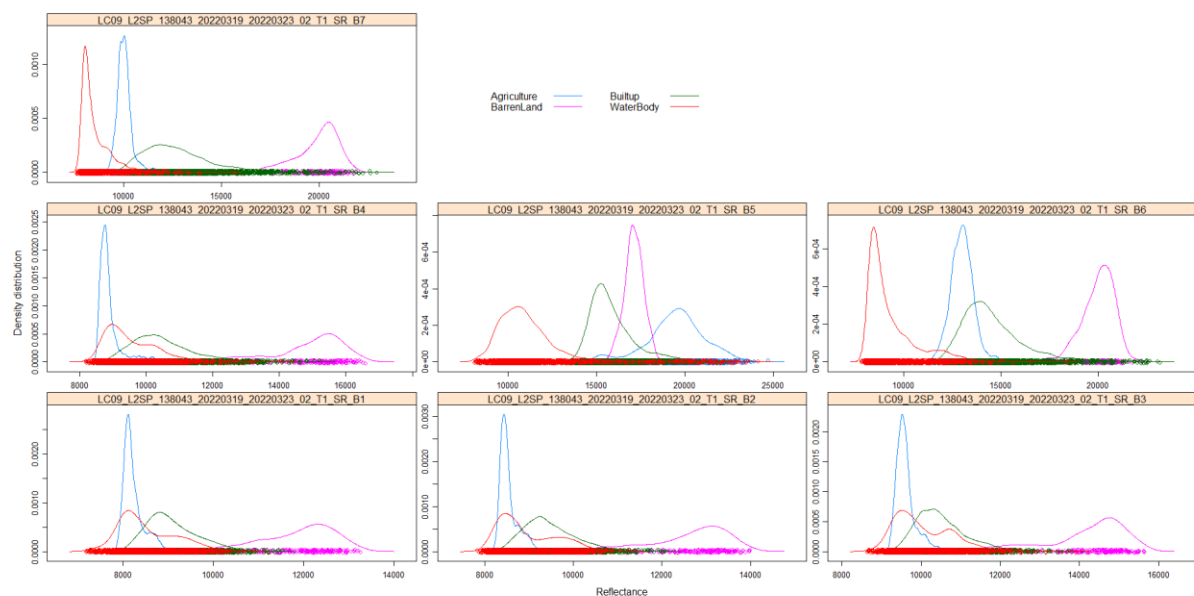


Figure 3.5. Band-wise Reflectance Distribution for Each Class

We can see, from the distribution of values. Almost all the curves are right skewed

Table 3.4. Skewness of the bands' distribution

Band Number	Skewness
Band 1	2.1014574
Band 2	1.8787203
Band 3	1.7342748
Band 4	-0.6004206
Band 6	0.9615210
Band 7	1.6587966

Only the distribution of Band 4 shows somewhat normal distribution. And the boxplot distribution shows lots of outliers in the training samples. Though it might seem tempting to remove the outliers from the samples, but as the number of classes are few and the spatial extent is quite large, those outliers may or may not contain important information for different regions.

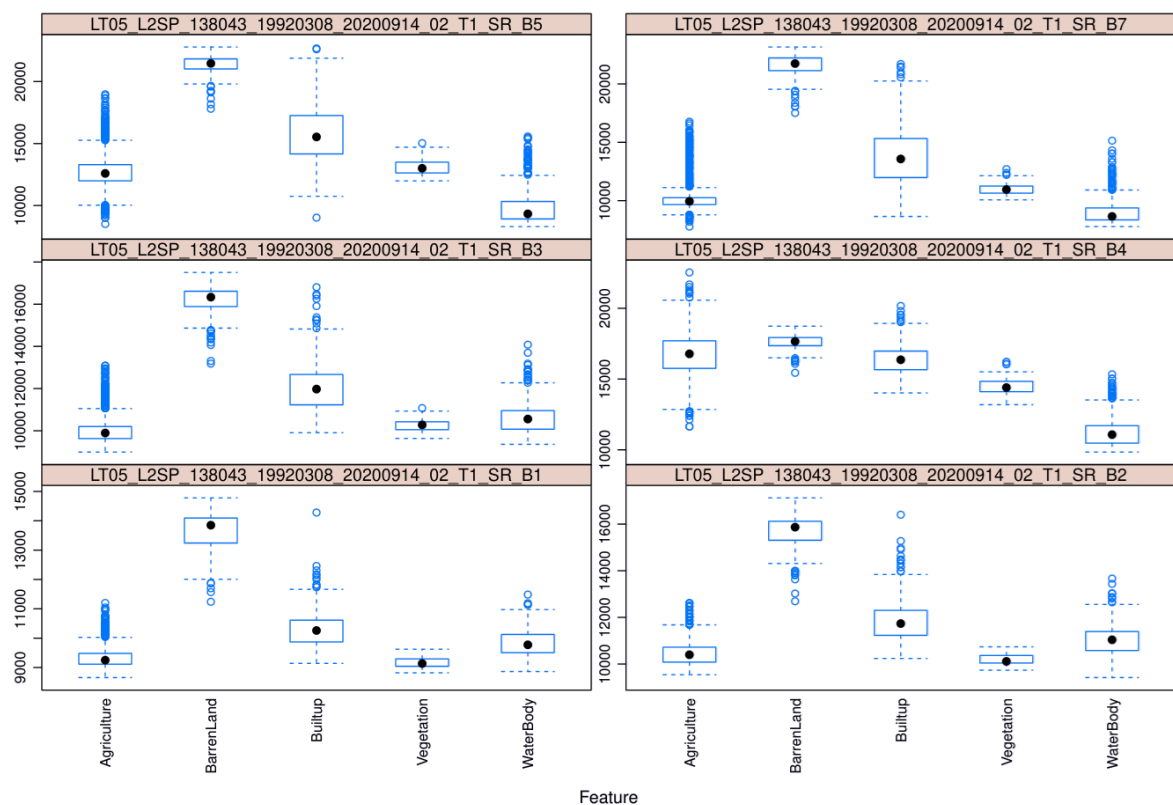


Figure 3.6. Whisker Diagram of Reflectance distribution across Bands

3.6.2 Employing Classification Models

Before confirming any particular model, different models were applied following their rule of thumb for each model as parameters using the training data from Padma region of the year 2010.

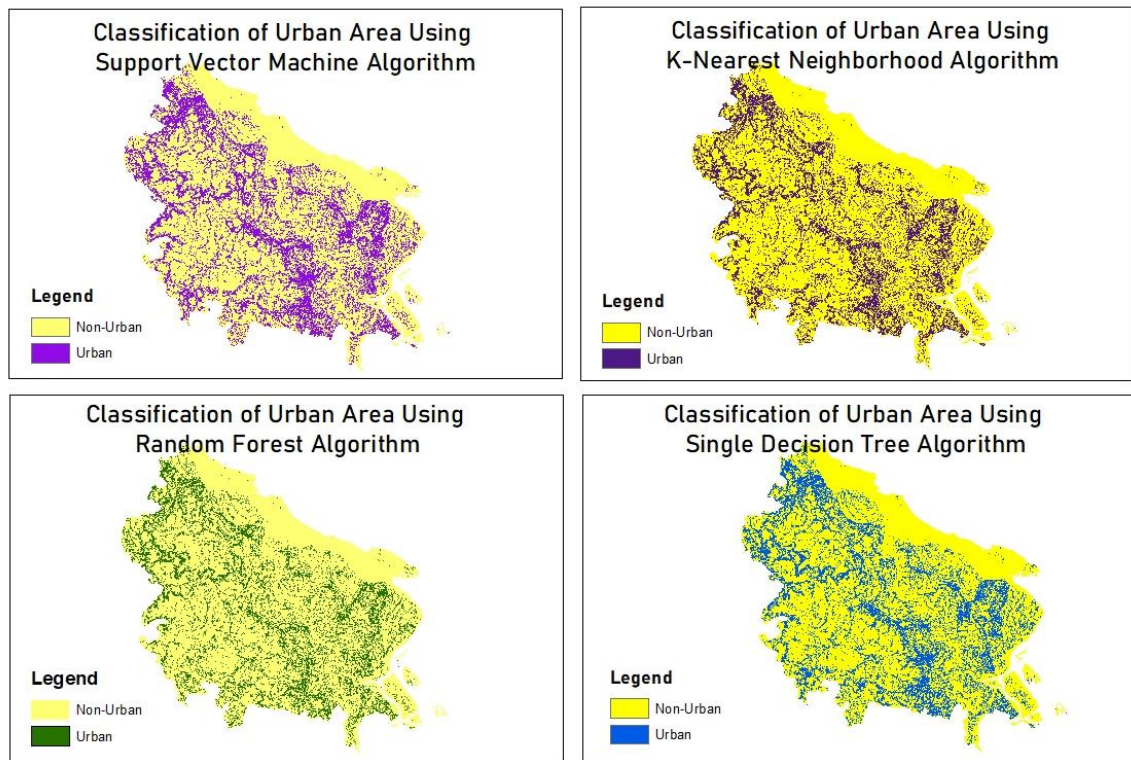


Figure 3.7. Output of different Methods

While observed in naked eye, all the models seem to provide similar results. Even when the overall accuracy is seen, all of them performed overwhelmingly better which is over 90% for all the cases.

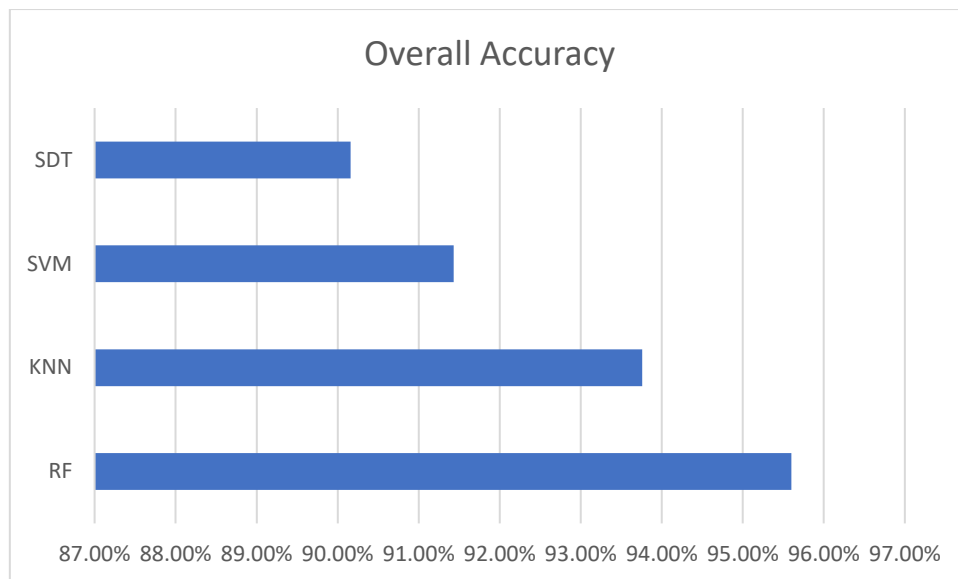


Figure 3.8. Overall accuracy of the applied models

Yet, Random Forest Performed slightly better than the other ones. Most of the classifier algorithms seem to overclassify urban areas where it is not in some areas. No matter what is the cause behind this is, it is clear that, random forest performed best among the tested classifiers. So, further into the research, random forest will be used as the classifier algorithm for rest of the cases.

Random Forest is a general term for ensemble methods using tree-type classifiers $\{h(x, H_k), k = 1, \dots, n\}$ where the $\{H_k\}$ are independent identically distributed random vectors and x is an input pattern. In training, the Random Forest algorithm creates multiple CART-like trees (Breiman et al., 1984), each trained on a bootstrapped sample of the original training data, and searches only across a randomly selected subset of the input variables to determine a split (for each node). For classification, each tree in the Random Forest casts a unit vote for the most popular class at input x . The output of the classifier is determined by a majority vote of the trees. The number of variables is a user-defined parameter (often said to be the only adjustable parameter in Random Forest), but the algorithm is not sensitive to it. Often, a blindly selected such value is set to the square root of the number of inputs. By limiting the number of variables used for

a split, the computational complexity of the algorithm is reduced, and the correlation between trees is also decreased. Finally, the trees in Random Forests are not pruned, further reducing the computational load. As a result, the Random Forest algorithm can handle high dimensional data and use many trees in the ensemble. This combined with the fact that the random selection of variables for a split seeks to minimize the correlation between the trees in the ensemble, results in error rates that have been compared to those of AdaBoost (Freund & Schapire, 1996) while being computationally much lighter. As each tree is only using a portion of the input variables in a Random Forest, the algorithm is considerably lighter than conventional bagging with a comparable tree-type classifier. The analysis of Random Forest (Breiman, 2001) shows that its computational time is $cT\sqrt{MN} \log(N)$ where c is a constant, T is the number of trees in the ensemble, M is the number of variables and N is the number of samples in the data set. It should be noted that although Random Forests are not computationally intensive, they require a fair amount of memory as they store an N by T matrixing memory. Generally, ensemble methods are black-box type classifiers. Of course, there are various methods that can be used to supplement them to gain more insight into the data. The nature of ensemble classifiers is such that much information can be garnered by clever use of bi-product data. To estimate the test set accuracy, the out of bag samples (the remaining training set samples that are not in the boot-strap for a particular tree) of each tree can be run down through the tree (cross-validation). The results are combined with a majority vote as before, and when compared to the actual labels, they yield a lower estimate of the classification error, as each sample can only be tested on the trees for which it was out-of-bag (so the effective ensemble for the out-of-bag estimate is smaller than the full ensemble used to train the entire forest). Similarly, the importance of variable m can be estimated by randomly permuting all the values of the m^{th} variable in the out of bag samples for each

classifier (thereby destroying the information contained in that variable). If an increased out-of-bag error is produced, that is an indication of the importance of that variable. Finally, one can obtain a measure of proximity between two samples, by counting for how many trees they end up in the same terminal node (this is normalized by dividing with the number of trees in the ensemble). When this measure is called for, a proximity matrix is computed. The proximity measure can take a significant amount of memory, NT , as discussed above. The proximity measure can for example be used to detect outliers. For each sample, the average squared proximity to the other samples in the class is calculated. The raw outlier measure is the number of samples divided by this average proximity (that is, the raw outlier measure will be large if the squared average proximity is small). The median and absolute deviation are calculated for the raw outlier measures of each class, and the raw measure for each sample in that class is then normalized by subtracting the median from the raw score and then dividing by the absolute deviation.

The number of decision trees for the models were set to 500 and adjusted the “mtry” parameter by testing multiple values and selecting the value that gave the best performance. Minimum number of samples per leaf were set to 5, which is a commonly used value for land cover classification problems. 70% of the training data were used for training the model and the other 30% were used to test the data the classified images were tested against the test data and the overall accuracy, kappa coefficient, and confusion matrix to evaluate the performance of the classifier. The overall accuracy obtained was between 87-97%, which indicated that the classifier was able to accurately classify the land cover classes in the study area.

Table 3.5.Accuracy of the classifications

Region	Sensitivity	Specificity	Balanced Accuracy	Kappa
Jamuna 1992	0.9366	0.9666	0.9516	0.8516
Jamuna 1998	0.9413	0.8935	0.9174	0.8274
Jamuna 2003	0.9533	0.9124	0.9329	0.8129
Jamuna 2009	0.8725	0.9036	0.8881	0.7381
Jamuna 2016	0.9636	0.9265	0.9460	0.8151
Jamuna 2022	0.9115	0.8637	0.8876	0.7776
Padma 2010	0.9448	0.9671	0.9560	0.8556
Padma 2014	0.8539	0.8995	0.8767	0.7867
Padma 2018	0.9215	0.9781	0.9498	0.8298
Padma 2022	0.9635	0.9832	0.9734	0.8334

3.6.3 Extracting Urban Area

As this study is focused on the urban growth prediction, other classes than the urban were reclassified into non-urban so regardless of what are the internal changes among non-urban classes are beyond the scope of the study. Another assumption for the classification for the study is no area that were previously urban will be converted into non-urban in future times. As the area that had these negative changes were minimal and generally is not an expected change, such areas were omitted in this study from ever being called urban.

In the observed 6 years urban area have increased from 1308sqkm to 1441sqkm. Arithmetically speaking, that is a 1.38% growth per year before construction of

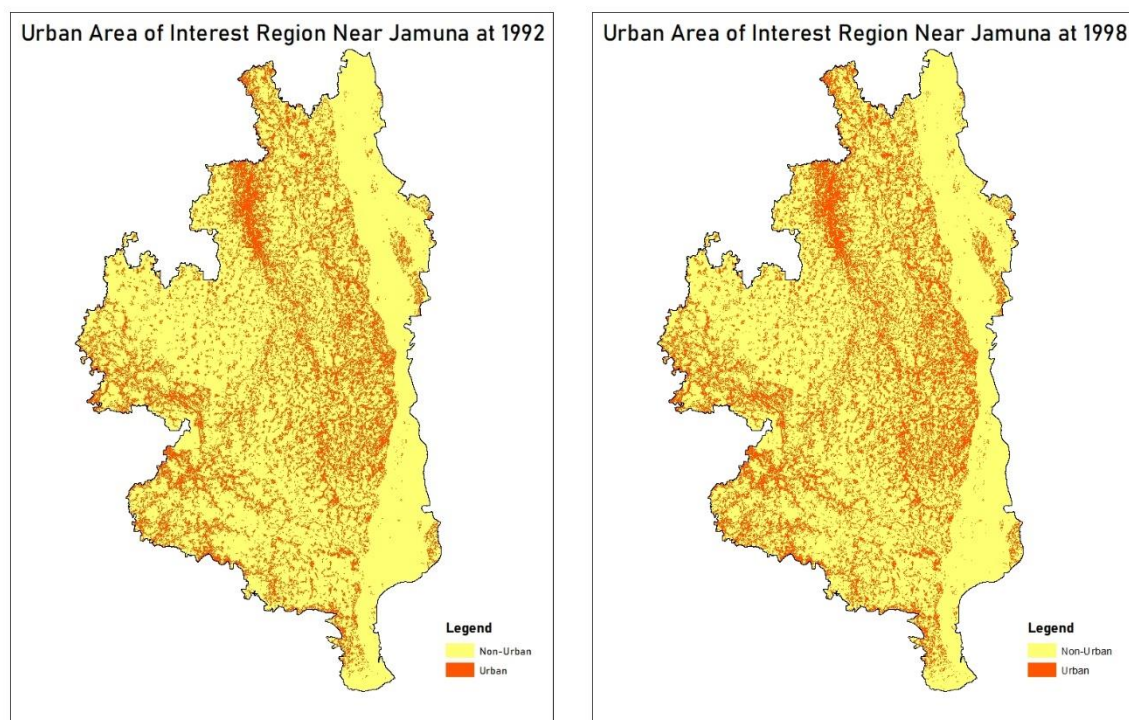


Figure 3.9. Urban condition of the Jamuna region before Jamuna Bridge

the Jamuna bridge. On the other hand, after the construction of the bridge in 6 years, in the year 2003, the urban share of that region was 1714sqkm which is

2.65% growth per year in arithmetic terms. It is quite clear the pattern of growth has shifted after the construction of the bridge.

Surprisingly throughout 24 years the growth rate was almost same till 2022. From

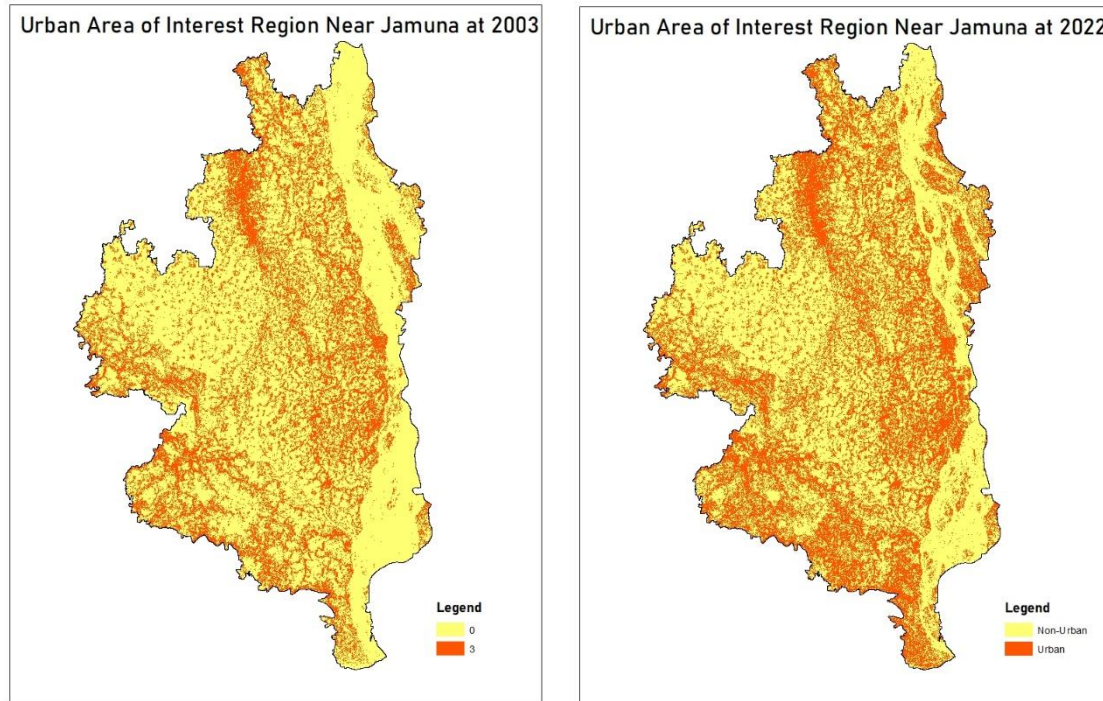


Figure 3.10. Urban condition of the Jamuna region after Jamuna Bridge

1998 to 2022 urban area has increased by 914 sqkm with a rate of 2.69% per year. Here one point to be noted that, 1% of growth in 1992 is not equal to 1% of growth in 2022 as urban growth are more prone to geometric form rather than arithmetic form.

Table 3.6. Urban changes in different timeframe

Region	Year	Urban Area(sqkm)	Growth(sqkm)	Rate
Jamuna	1992	1308	-	-
Jamuna	1998	1417	109	08.33%
Jamuna	2003	1580	163	11.50%
Jamuna	2009	1787	207	13.10%

Jamuna	2016	2032	245	13.71%
Jamuna	2022	2331	299	14.72%
Padma	2010	1368	-	-
Padma	2014	1537	169	12.35%
Padma	2018	1722	185	12.03%
Padma	2022	1934	212	12.31%

3.7 Predicting Urban Growth

3.7.1 Prediction model

For this study, growth prediction is done by the model provided by DINAMICA EGO. The model consists of 5 steps to form a transition probability of a region and giving a prediction map of the urban state of the given region (Soares-Filho et al., 2002).

Dinamica EGO model calculates land use transition matrix and cell transition probability by calling a series of built-in functions, and simulates the final landscape of termination year according to specific conversion rules. The modeling process is shown in.

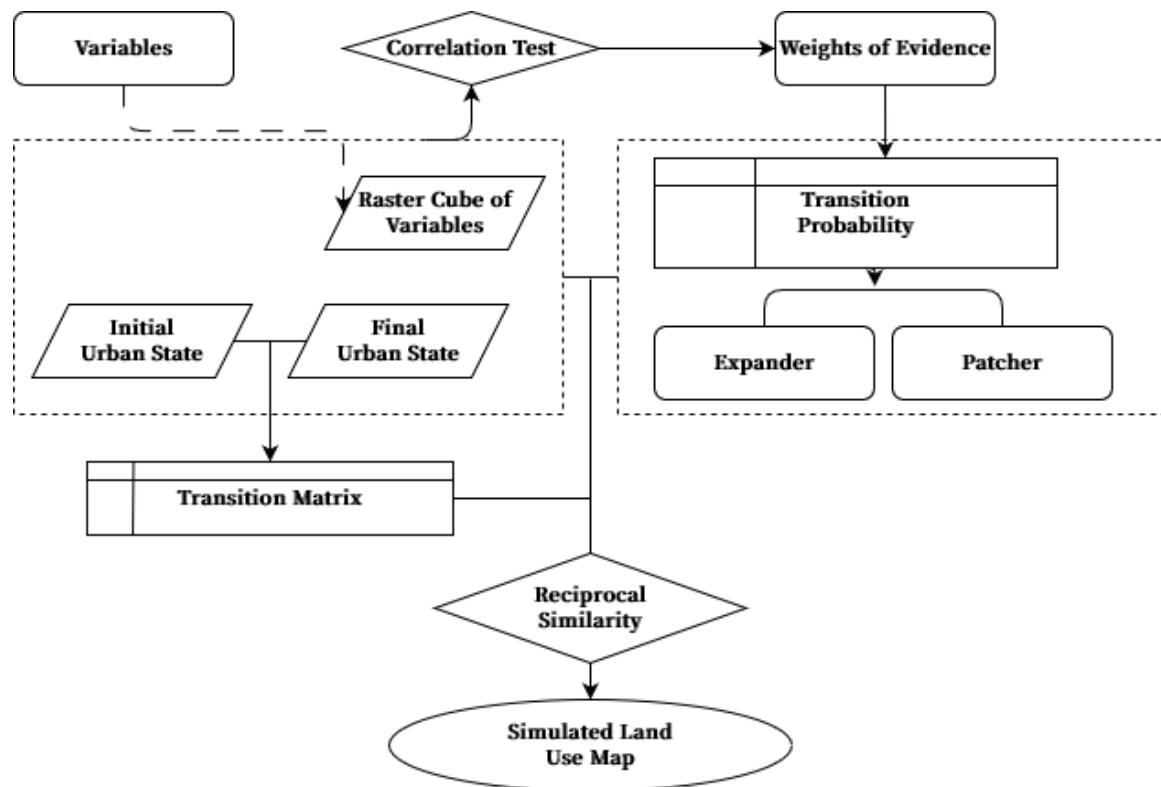


Figure 3.11. Growth prediction model flow chart

The key steps are as follows:

1) Calculating land use transition matrix, which is a square matrix describing the transitions of each state in a system over a given period. The matrix is created using the Markov Chain Method. The transition volumes are the sum of fixed percentages of all variables in previous time step. Single-step matrix, is the ratio of change amount of land use to the number of land use in initial year. Dividing the process into multiple time steps, n , the relation between multiple-step matrix and P can be expressed as:

$$P = (P_t)^n$$

In the simulation process, more than one multiple-step matrix can be set according to the adjustment of policy or other factors in different time periods to control the land use transition rates. In this study, multiple step matrix is used and each step corresponds to one year.

2) Calculate transition probability, which describes the probability that the driving factor will cause land use type to change. Some driving factors change when cell states change, called dynamic variables, and those that do not change with the iterative process are called static variables. Dinamica EGO model receives a raster cube composed of driving factors as input.

Dinamica EGO model uses the Weights of Evidence (WoE) method based on Bayes criterion to calculate transition probability. The possibility of a certain land use conversion is represented by the posterior probability for land use change to occur under the condition of combined driving factors (Agterberg & Cheng, 2002). Dividing study area into $N(T)$ cell units, defining D as the event of land use change to be predicted, whose number of cell units is $N(D)$, the prior probability of land use change can be calculated as follows:

$$P(D) = \frac{N(D)}{N(T)}$$

The odds ratio $O(D)$ is given by $O(D) = \frac{P(D)}{1-P(D)}$.

The driving factors of land use changes in the study area are regarded as evidence factors, and indicates the event that a certain evidence factor exists. The weighting factor W^+ is described as follows:

$$W^+ = \left[\ln \frac{P\left(\frac{B}{D}\right)}{P\left(\frac{B}{D'}\right)} \right]$$

$$W^- = \left[\ln \frac{P\left(\frac{B'}{D}\right)}{P\left(\frac{B'}{D'}\right)} \right]$$

D' represents land use change to be predicted does not occur and B represents the evidence factor is absent. The contrast (C) is $W^+ - W^-$, which measures association between the occurrence of event D and factor C . When C is close to zero, the factor has no meaning for land use change. If there are m evidence

factors, which can be labelled as $B_1, B_2, \dots, B_m$, affecting the occurrence of event D and all independent, then,

$$\ln O(D|B_1 \cap B_2 \dots \cap B_m) = \sum W^+ + \ln O(D)$$

The posterior probability of a transition of land use type I to j under all evidence factors at location (x, y) is:

$$P(i \Rightarrow j | B_1 \cap B_2 \dots B_m) = (e^{\sum W^+}) / (1 + e^{\sum W^+})$$

When calculating the weight coefficients, WoE method requires driving factors are discretized. Dinamica EGO uses the method developed by Agterberg and Bonham-Carter to discretize continuous variables such as elevation, slope, distance, etc. It assumes that the spatial correlation of driving factors at close range are similar, and the weight of each interval indicates the impact of driving factors on land use changes in different areas (Carranza-García et al., 2019). The Crammer test and the Joint-Uncertainty Information are used to assess whether each driving factor is independent from any others which is required by WoE (Bonham-Carter & Bonham-Carter, 1994).

3) Set transition rules to define how cell states are updated over time. In modeling process, cell growth pattern is mainly determined by cell diffusion rule, patch size, patch standard deviation and isometric coefficient et al. There are two cell transitional functions, Expander and Patcher (Chadid et al., 2015). The Expander function expands or contracts same land use types based on the original cell. Detecting edge cells as extended objects, it decides whether to spread or not according to the state of its neighbor cells. The Patcher function selects a new cell as the core and spreads a specific number of cells to form a new patch. Users determine the ways of all cell transitions by setting the ratio of Expander to Patcher. The ratio of Expander to Patcher is equal to the ratio of the number of cells in new patches which have a public edge with previous existing land use patches to that not have. This study used Kappa coefficient to assess the simulation results obtained by the Dinamica EGO model, which initially is a

statistic for discrete multivariate analysis (Pontius Jr & Schneider, 2001). The expression of the Kappa coefficient can be generally expressed as follows

$$\kappa = \frac{P_0 - P_c}{P_p - P_c}$$

Where P_0 represents the observed correct proportion, P_c represents the expected correct proportion, and P_p represents correct proportion in an ideal scenario. The value of Kappa confidence lies within the interval [0, 1], and closer to 1 represents a higher similarity. As this study is done on finer images with 30m resolution, gaining higher prediction accuracy with that precision is almost impossible. In that precision telling where the growth will take place will be like showing the location where a person will build their house which is virtually impossible with high accuracy. So, in this study, a fuzzy similarity method is applied where the model will iterate by increasing radius to match similarity. The maximum similarity value shows the accuracy when the maximum value is taken from the fuzzy chart. Minimum similarity value gives the accuracy when the minimum value is taken. Maximum similarity is more effective when it is a comparison between land class vs land class while minimum is good for literal map comparison. And we can check within what radius the model is performing better. For example, with constant decay function, an accuracy of 50% using 21 windows will be equivalent to an accuracy of 50% within 300m radius. Meaning, if there is a predicted growth within a radius of roughly 300m where there is an actual growth, the model will mark that as correct prediction. And 50% accuracy means the model have predicted 50% of growth correctly in that range.

3.7.2 Creating Growth Models

In this study, nine models were created to predict the impact of the Padma bridge. The models are:

- 1. Pre Jamuna-Bridge Model:** This model helps us understand how growths were taking place in Jamuna region before the bridge was built. This model will help us finding the difference between the Jamuna and Padma region without the bridge.
- 2. Post Jamuna-Bridge BAU Model:** This model will show what will be the scenario if Jamuna bridge was not built using the transition and weight values from Pre Jamuna-Bridge Model.
- 3. Post Jamuna-Bridge Model:** This model will help to understand how Jamuna Bridge impacted urban growth while compared with the Post Jamuna-Bridge BAU model.
- 4. Adjusted Pre Jamuna-Bridge model for Post Jamuna-Bridge Scenario:** This is the adjusted pre Jamuna bridge model where the transition and Weights were adjusted using the post Jamuna model. This will give us an idea of how the adjusted models work.
- 5. Pre Padma-Bridge Model:** This model shows how urban growth is taking place in the Padma region.
- 6. Pre Jamuna-Bridge Model for Pre Padma-Bridge Scenario:** This model will help to check the context difference between the two regions. It will show how well would pre Jamuna-Bridge model perform for Pre Padma-Bridge scenario.
- 7. Adjusted Pre Jamuna-Bridge model for Pre Padma-Bridge Scenario:** This is the adjusted model of Pre Jamuna bridge using the difference between Model 5 and Model 6. This will help to understand how the context adjusted models will perform.
- 8. Adjusted Post Jamuna-Bridge model for Post Padma-Bridge Scenario:** This is the final model which will show how urban growth will take place in Post Padma-Bridge scenario at 2046. It will be consisted of the Pre Padma-Bridge model adjusted with Jamuna bridge impact values (used in Model 4) and the context adjustment values (used in Model 7).

9. Post Padma-Bridge BAU Model: This model shows what will be the urban growth condition if the Padma bridge was not built. It will give the urban landscape of 2046 with current growth rate and transition probability.

Chapter 4: Analysis

4.1 Outcomes from the models

Pre Jamuna-Bridge Model was trained using growth of Jamuna region from 1992 to 1998. Primary objective of this model was to finding difference between post construction of the bridge versus pre construction of the bridge growth prediction. Pre Jamuna bridge model performed with an average accuracy of 80% within a 25x25 cell window. There accuracy assessment done in reciprocal similarity method, one is

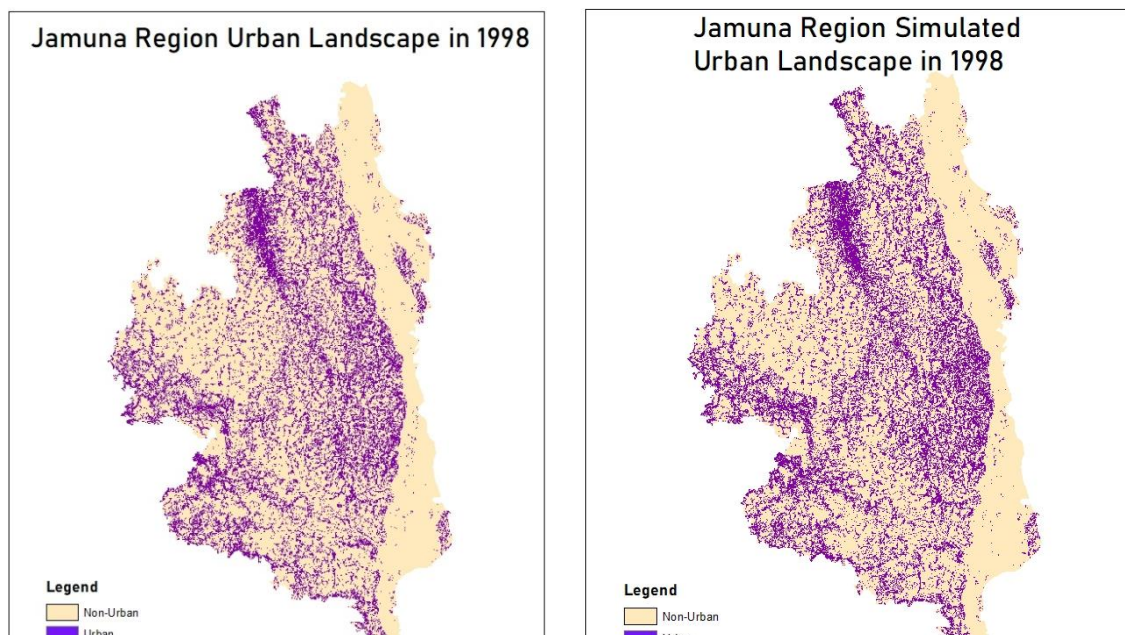


Figure 4.1. Comparison of real scenario and Model 1 simulation.

done assessing the prediction against the real condition other is assessing the real condition against the prediction. The value which is higher in similarity is considered the maximum value of similarity and the minimum is set as the minimum value of similarity. When considered the “maximum value of similarity”, the model performed almost 100% accurately within a 17x17 window. When considered “minimum similarity”, the model predicts more than 60% growth in a 25 x 25 cell radius. This model shows high accuracy as expected as it is trained from the real scenario and no adjustments were made.

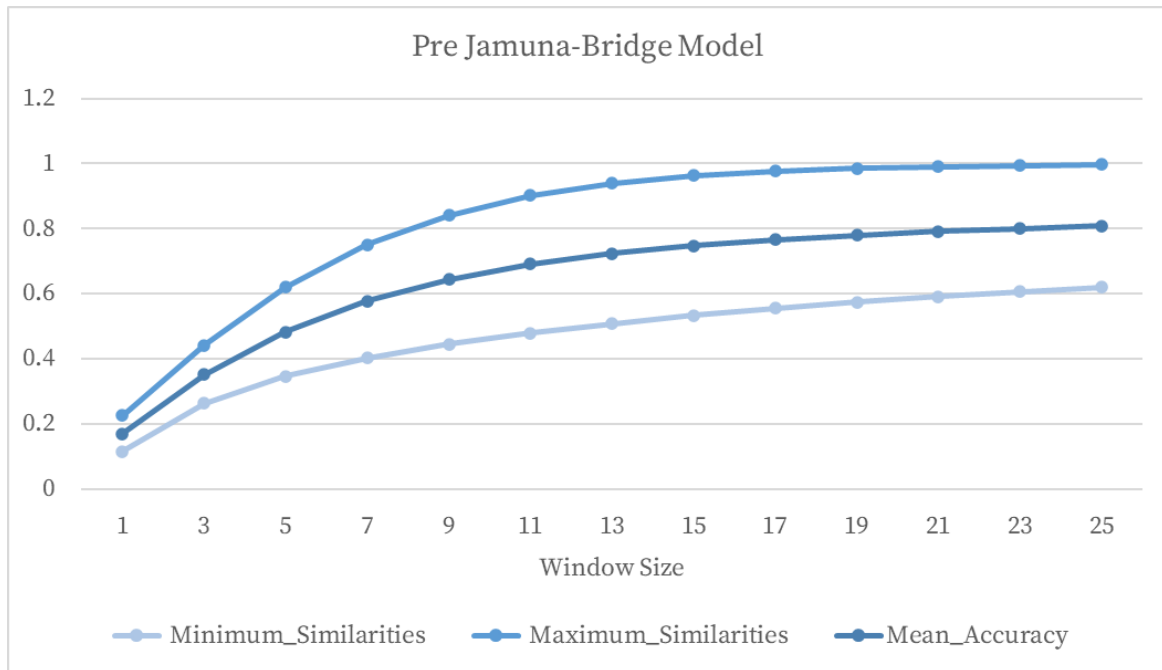


Figure 4.2. Similarity graph based on window size for model 1

for **Post Jamuna-Bridge BAU Model** transition matrix and transition probability values were used from previous model but initial landscape was used of the urban form of 1998 at the Jamuna bridge region. This model shows what would be the

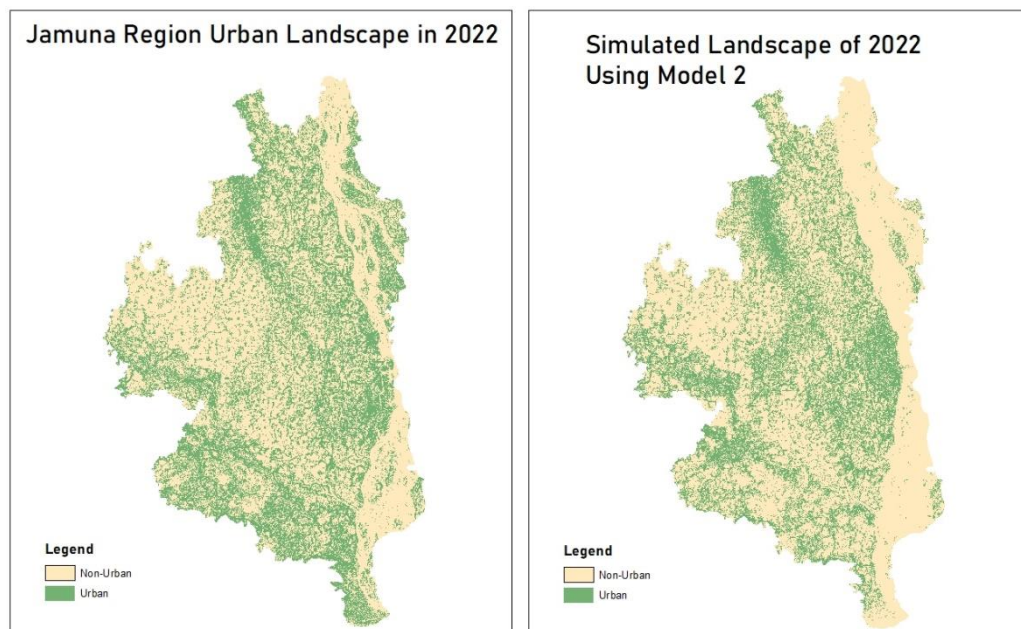


Figure 4.3. Comparison of real scenario and model 2.

landscape of that region in 2022 if the growth pattern followed the previous trend where Jamuna bridge still was not built.

This model also shows quite promising result but not as good as the previous one. Within a 25x25 window the maimum accuracy is nearly 85% where in that radius the

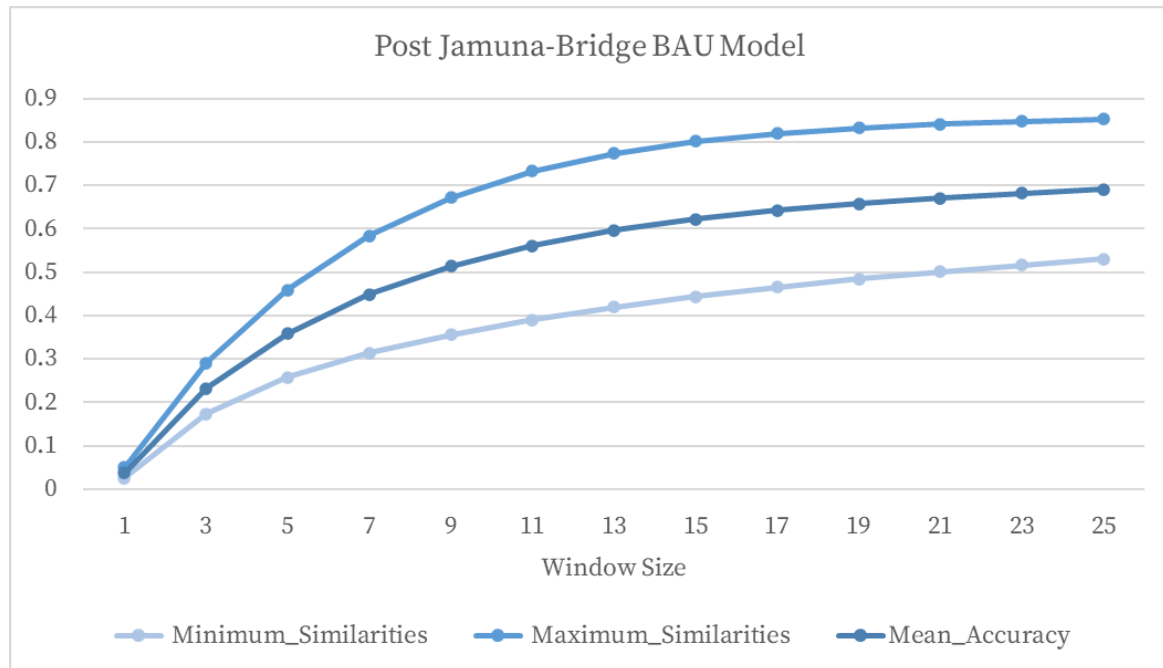


Figure 4.4. Similarity graph based on window size for model 2

previous model was almost 100% the average accuracy is almost 70% and minimum similarity value is roughly above 50%.

Post Jamuna-Bridge Model uses the pattern of growth after Jamuna bridge was constructed. This simulation shows rather high cluster of growth but matches the growth cell counts.

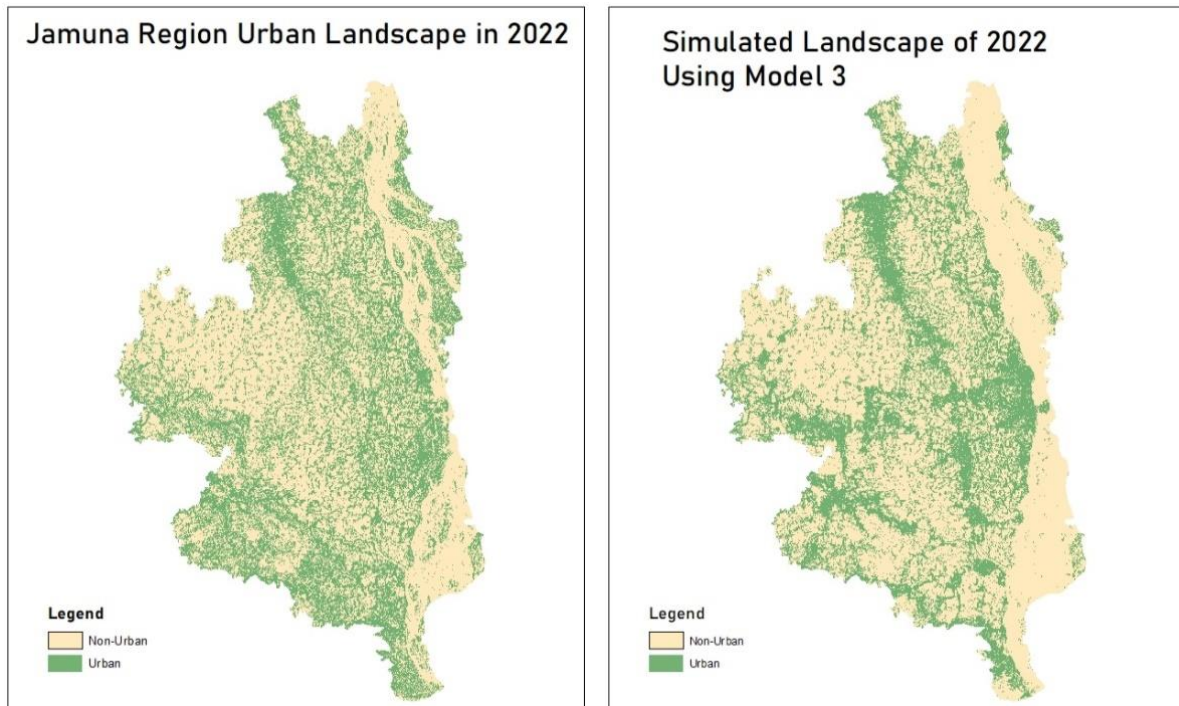


Figure 4.5. Comparison of real scenario and Model 3

There is low difference between the minimum value similarity and maximum value similarity when the window size is low, but as the window size increased, the model was proven to be fairly accurate. With a maximum accuracy close to 100% when the window size increased. One thing to note here, is visual similarity contains fewer weight here than the number of cells changed to growth within a vicinity as constant decay function is being used.

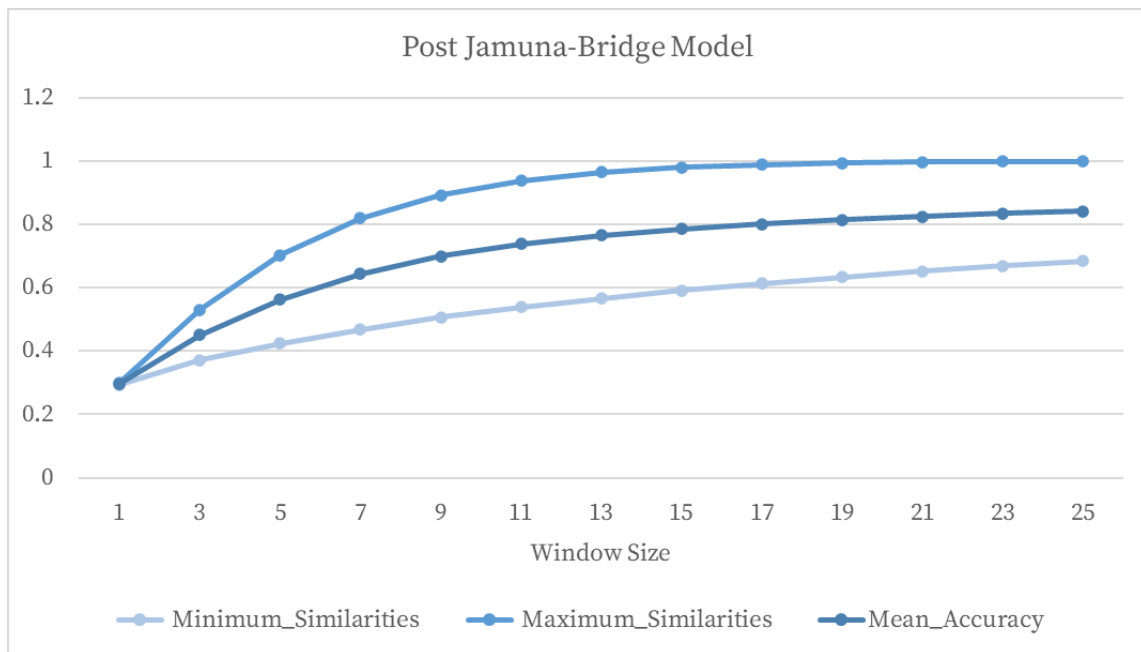


Figure 4.6. Similarity graph based on window size for model 3

in ***Adjusted Pre Jamuna-Bridge model for Post Jamuna-Bridge Scenario*** model the transition and weight differences were adjusted into the matrix. The outcome shows

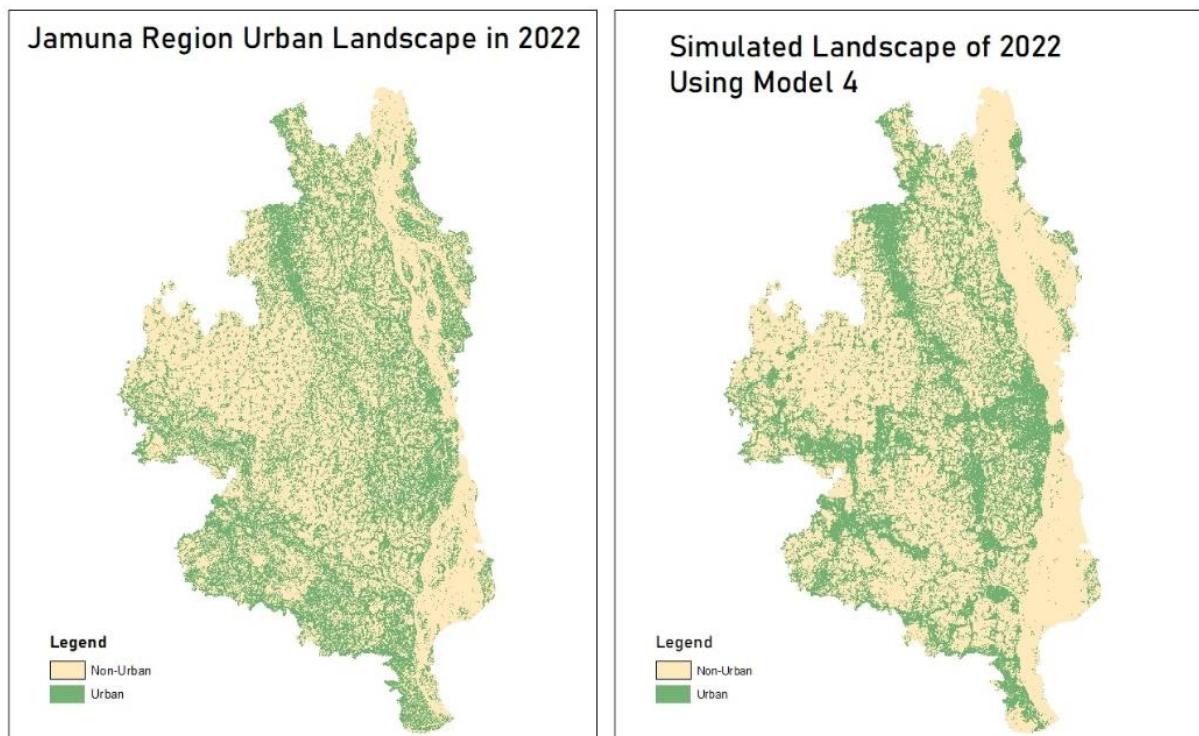


Figure 4.7. Comparison of real scenario and Model 4

almost similar results as the original post Jamuna model. This similarity assures that, adjustment is not causing any severe damage to the model integrity.

The similarity graph also shows almost similar result as Model 3. So, the adjusted

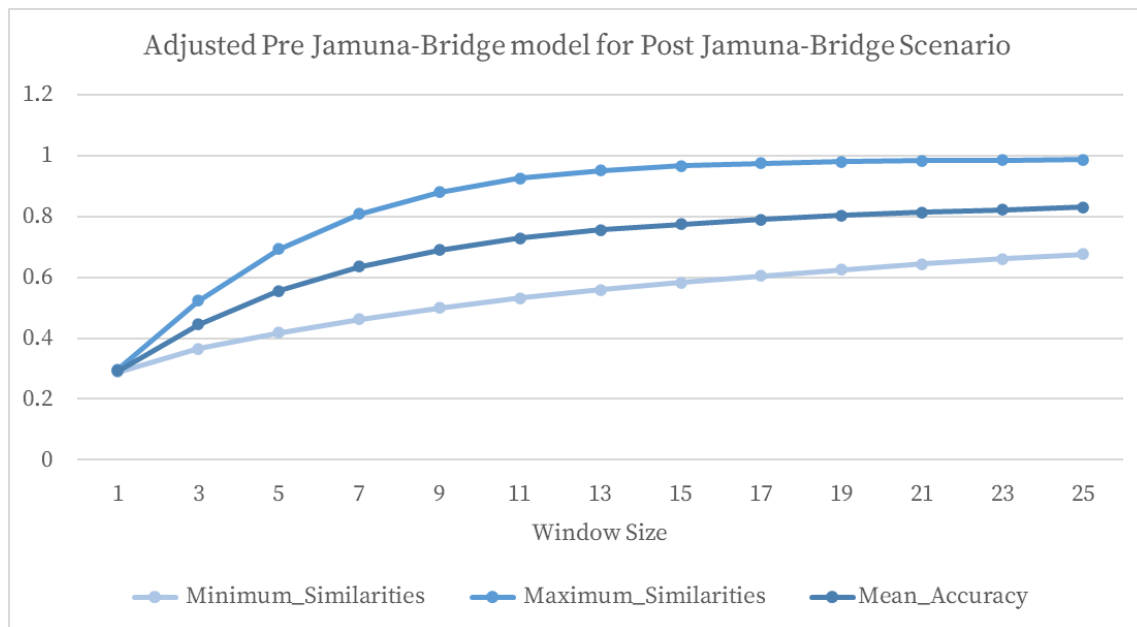


Figure 4.8. Similarity graph based on window size for model 4

model produces almost similar result as the original model. As the data was adjusted The **Pre Padma-Bridge Model** is built on the changes in urban growth from 2010 to 2018 in the Padma region. With these data Urban landscape is predicted of 2022.

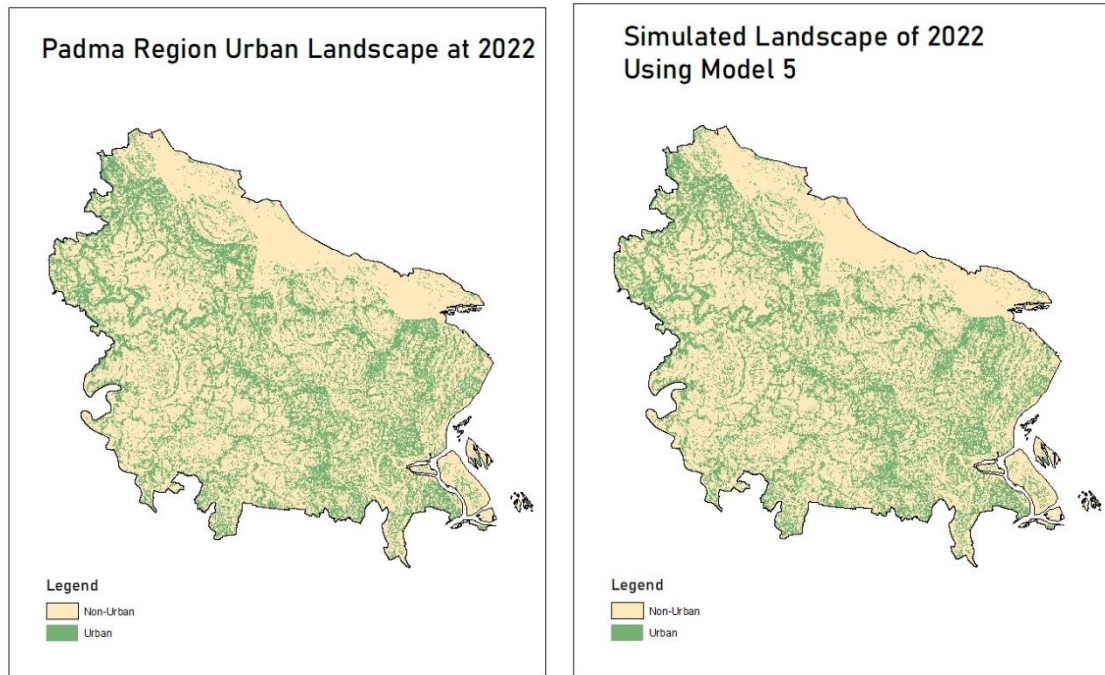


Figure 4.10. Comparison of real scenario and Model 5

In the similarity assesment, the pre Padma model performed outstandingly. With a mean accuracy of over than 90%, it has performed very well even with lower window dimensions. With a window size of 9x9 it has an average accuracy of almost 80%.

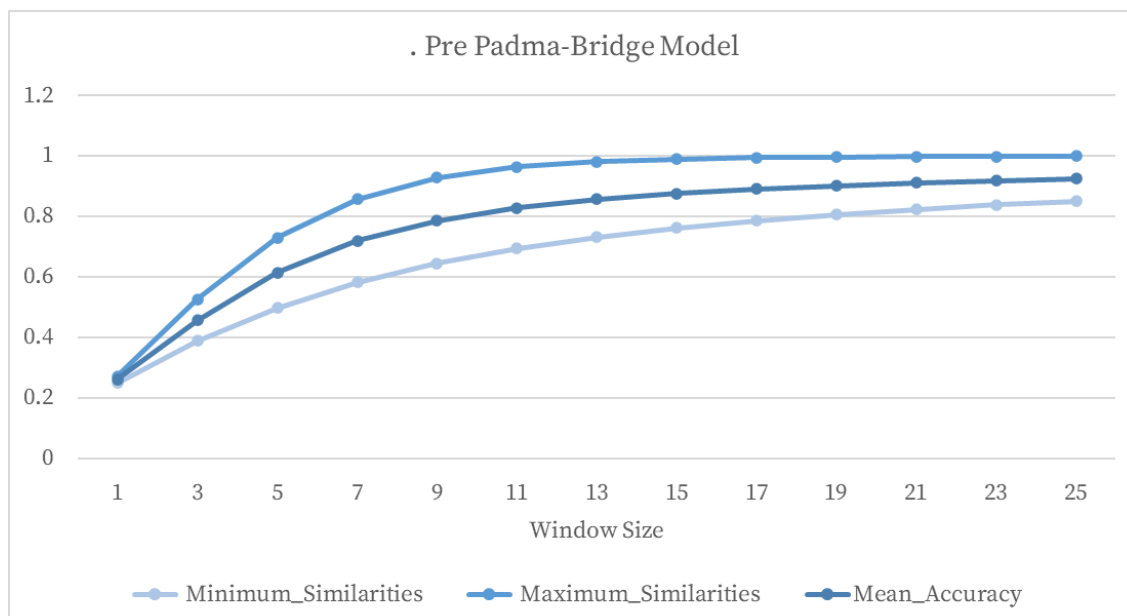


Figure 4.9. Similarity graph based on window size for model 5

The *Pre Jamuna-Bridge Model for Pre Padma-Bridge Scenario* was built with the unchanged transition properties derived from the pre Jamuna model. Finding the

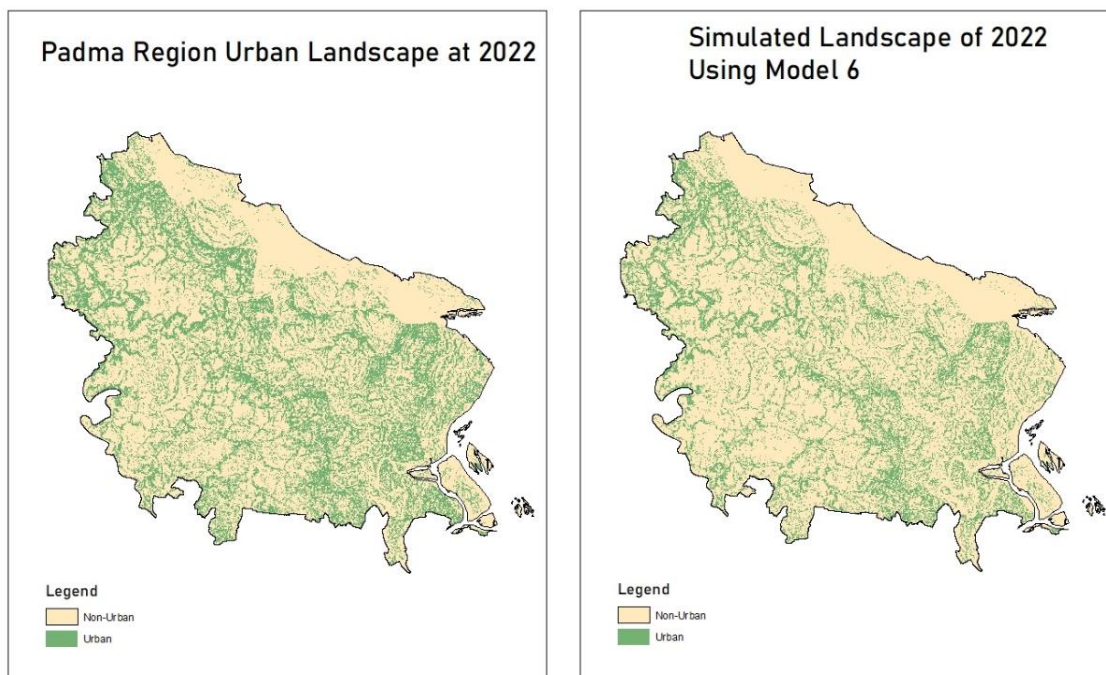


Figure 4.11. Comparison of real scenario and Model 6

difference between these two models will be provide with the context adjustment value for the Jamuna region and Padma region.

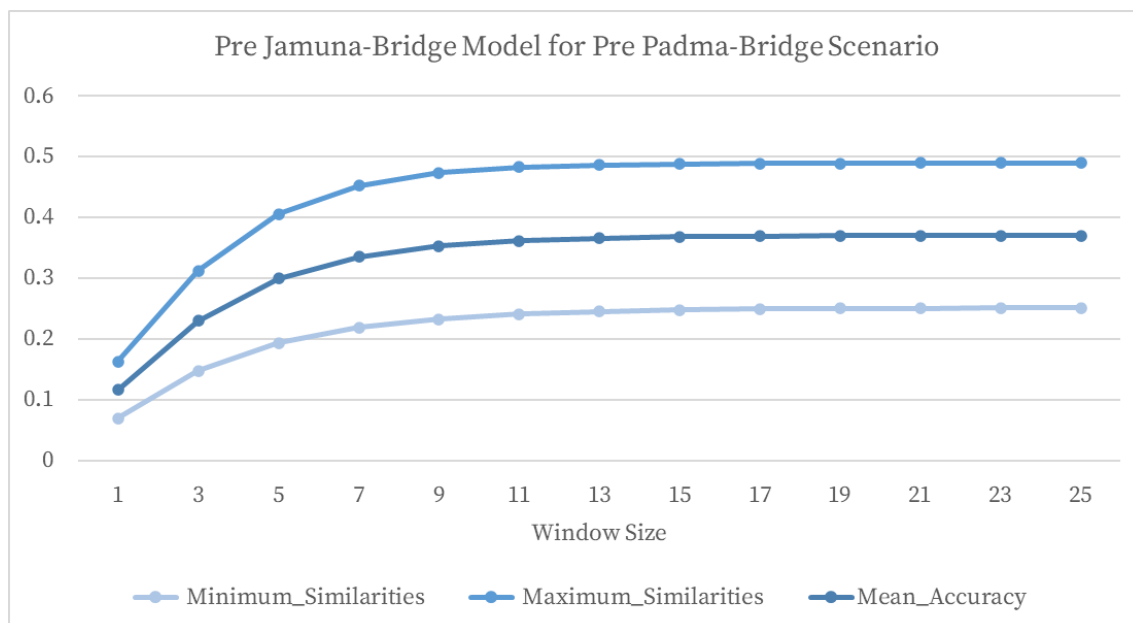


Figure 4.12. Similarity graph based on window size for model 6

As seen from the graph, this model has not performed quite up to the mark. Even after the 9x9 window size, increment in window size could not help. Although this bad outcome was expected as the used values are from a different model and a different timespan. Now, using the difference of this model with model 5, model 7 was created which is a pre Jamuna adjusted model for predicting pre urban growth of Padma region. If that model performs considerably well, it will be justifiable to say that, post Jamuna will do the same.

The ***Adjusted Pre Jamuna-Bridge model for Pre Padma-Bridge Scenario*** model is derived from the pre Jamuna model in which, the transition values were adjusted with the difference values of the model 5 and 6.

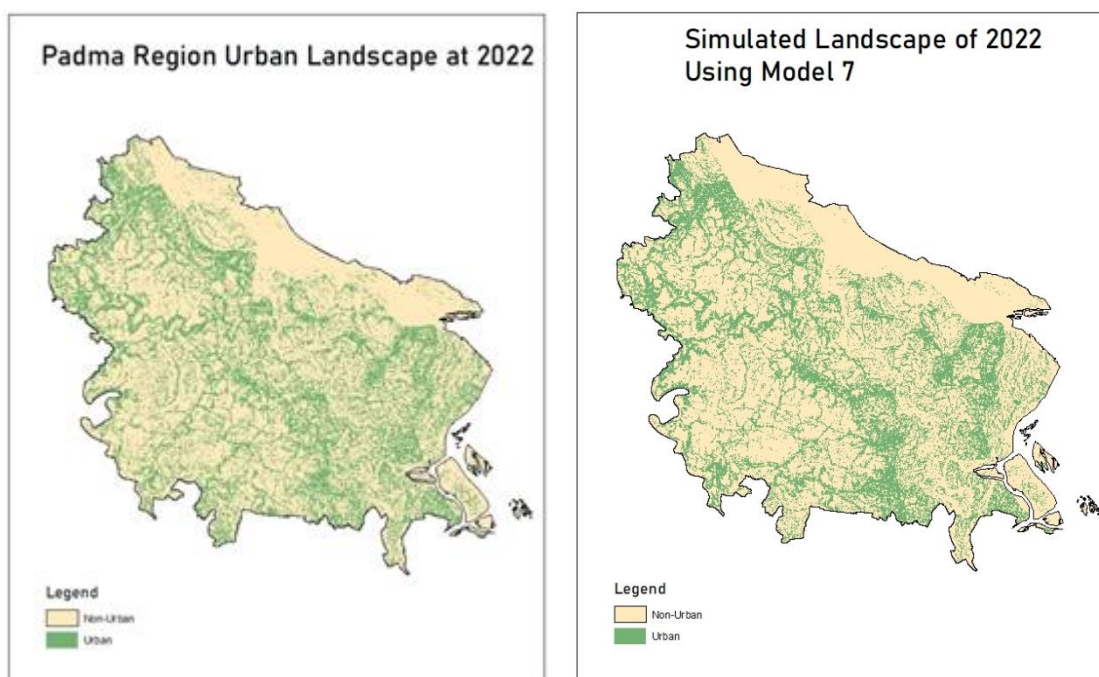


Figure 4.13. Comparison of real scenario and Model 7

As seen from the following graph,

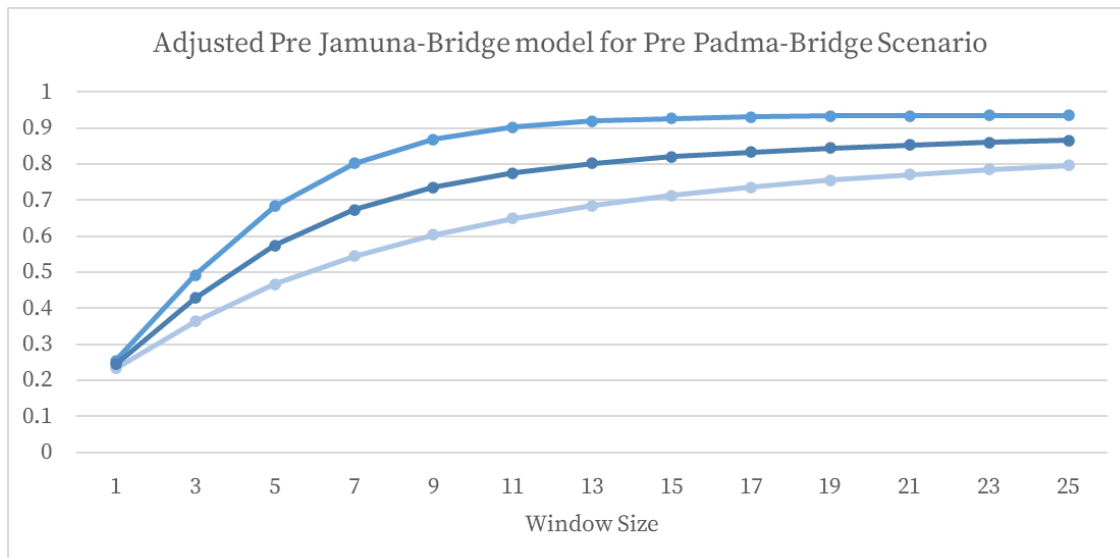


Figure 4.14. Similarity graph based on window size for model 7

The adjusted model performed exceptionally well considering the facts that it was not trained directly through the specific context. With an average accuracy of more than 85%, at a window size of 25x25, it can safely be said that, adjusting the model with difference adjustment within these two contexts, works fine enough. There is around 5-6% growth that was unexplained with the adjusted model. If seen closely the growth pattern in the char region is different for Padma River than Jamuna. There is a notable amount of growth in the chars of Jamuna river and there is a good number of pocket chars which are smaller in size and completely enclosed by wider areas of river and far from the riverside locality. The situation is quite different for the Padma region. Livelihood styles are also different in these two regions. Even the char people are not dependent that much on fisheries in Jamuna region where there is a significant population in Padma region depends on fisheries. That makes spatial correlation between growth and distance from waterbodies slightly different in these regions. Still the unexplained portion is quite minimal.

Adjusted Post Jamuna-Bridge model for Post Padma-Bridge Scenario is a model where we will consider that the Padma bridge will have a similar impact like the Jamuna bridge. But the impact will be adjusted to the context. For example, if nobody built their houses near the National highway in Jamuna region but Proximity to National highway is a key variable of interest in Padma region, the model will incorporate and enhance the role of National highway in this region.

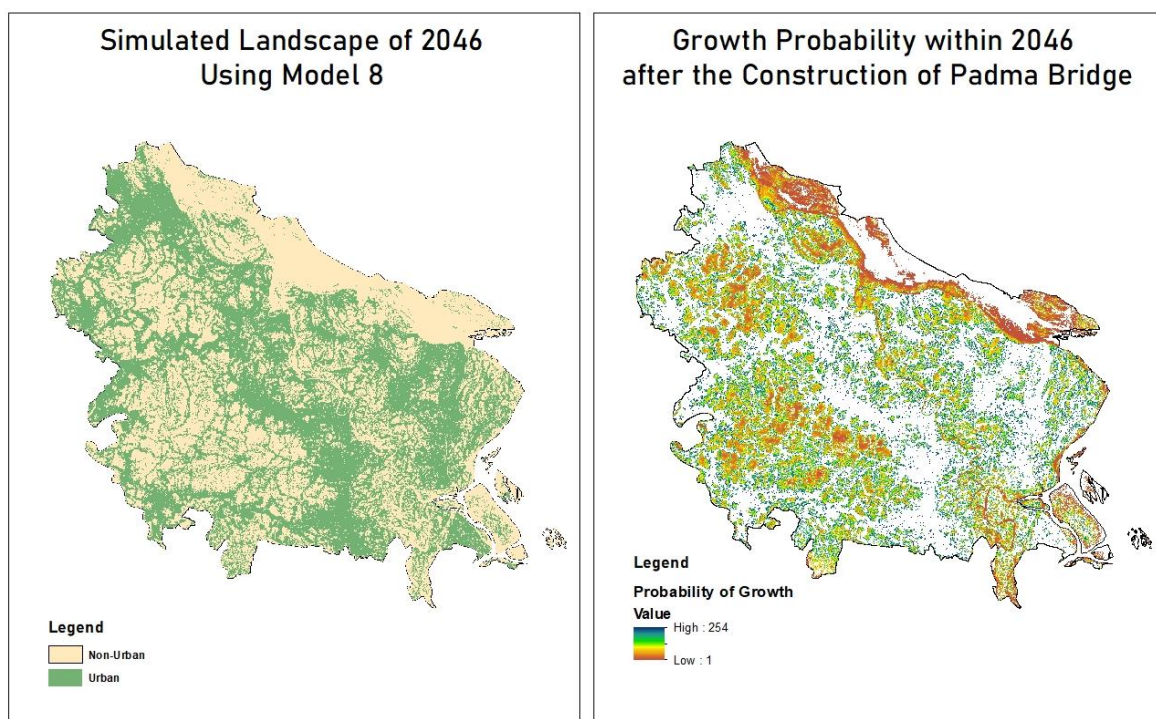


Figure 4.15. Predicted Growth of 2046 and Transition probability

This, is the 24-year prediction of the Padma Region considering the impact of the Padma bridge. To extract the locations of impact, another prediction is needed to be made where the impact of Padma bridge will not be considered. So, using the values from model 5, a 24-year model were also run to check how would be the scenario, if the Padma bridge were not built. The ninth model named ***Post Padma-Bridge BAU Model*** simulates that.

As we can see, the impact of the bridge is spread all over the region. Even if the result may seem counterintuitive that, why the construction of an infrastructure in a single

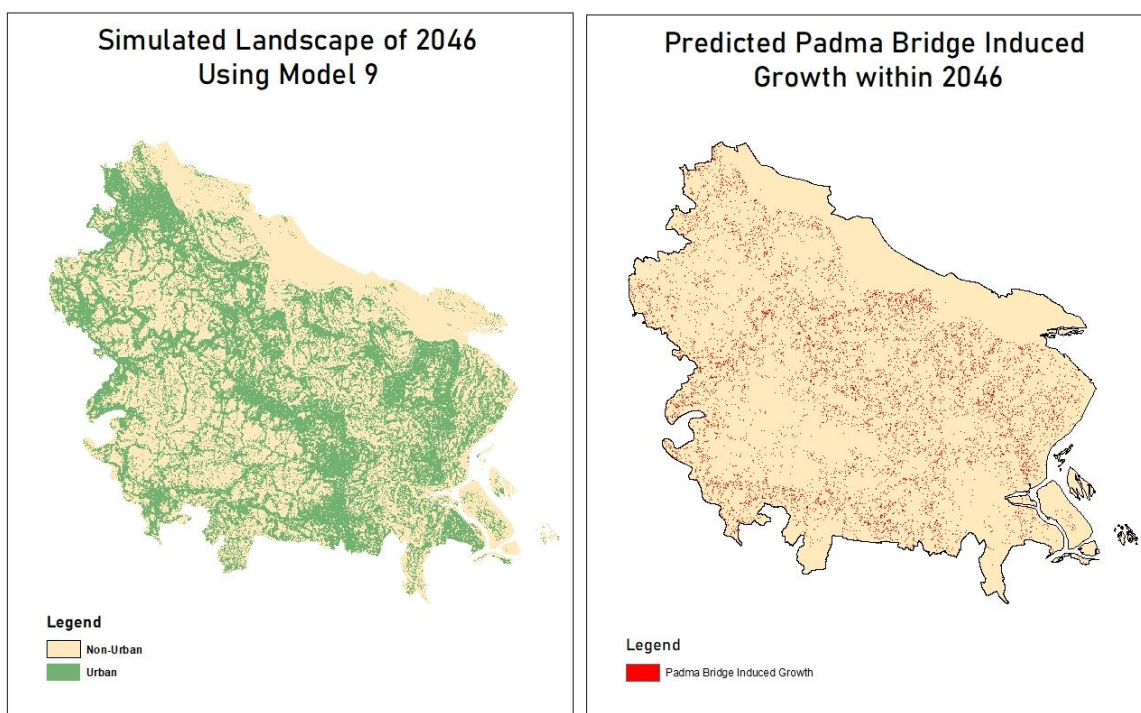


Figure 4.16. Predicted impact growth of Padma bridge

place will put such impact in a distant region than the closer ones.

Again, the bridge isn't supposed to cause growth on its own rather it is expected to enhance the conditions that already causes growth. So, it is expected to have more growth where there is an amicable condition for growth already. Approximately additional 317sqkm area of urban growth will be influenced by the Padma bridge.

Following map helps finding how urban growth in each Upazila is influenced by the Padma Bridge



Figure 4.17. Upazila-wise Padma Bridge influenced Predicted growth within 2046

Here, it is clear, not all upazila are equally influenced by the bridge. Gopalganj Sadar and Madaripur Sadar are the most influenced upazilas where upazilas like Hizla, Muladi or Char Bhadrasan are not that much influenced by the bridge.

Chapter 5: Conclusion

5.1 Conclusion

This study aimed to develop a new model for predicting the influence of the Padma bridge on urban growth on its surrounding area. Through remote sensing, GIS based approaches and machine learning the study was able to predict that the Padma bridge will influence approximately 317 sqKm of urban growth if it keeps a similar impact like the Jamuna Bridge.

The study has also provided with a probability map to assess where it is most likely to form new growth and which upazilas will need more attention to facilitate upcoming growth. This study can help the planners to create longitudinal plans regarding the area.

The study also has some limitations, socio-economic variables play an important role on urban growth but could not be incorporated into the study due to lack of availability in different timespan. Also, this study assumed the Padma bridge will put similar impact like Jamuna bridge, but the extents were unknown.

Overall, this study has made a significant contribution to the field of urban growth prediction, and detecting influence of a newly built infrastructure on urban growth. The models developed in this study also have the potential to be executed beyond the scope of this study.

References

- Abbas, A. W., Minallh, N., Ahmad, N., Abid, S. A. R., & Khan, M. A. A. (n.d.). *SI NDH UNIVERSITYRESEARCH JOURNAL (SCIENCESERIES)*. 4.
- Agterberg, F. P., & Cheng, Q. (2002). Conditional independence test for weights-of-evidence modeling. *Natural Resources Research*, 11, 249–255.
- Aisa, B., Mingus, B., & O'Reilly, R. (2008). The emergent neural modeling system. *Neural Networks*, 21(8), 1146–1152.
- Batty, M. (2005). Agents, cells, and cities: New representational models for simulating multiscale urban dynamics. *Environment and Planning A*, 37(8), 1373–1394.
- Berberoglu, S., Lloyd, C. D., Atkinson, P. M., & Curran, P. J. (2000). The integration of spectral and textural information using neural networks for land cover mapping in the Mediterranean. *Computers & Geosciences*, 26(4), 385–396.
- Bonham-Carter, G. F., & Bonham-Carter, G. (1994). *Geographic information systems for geoscientists: Modelling with GIS*. Elsevier.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- Breiman, L., Friedman, J., Olshen, R., & Stone, C. (1984). *Cart. Classification and Regression Trees*.
- Carranza-García, M., García-Gutiérrez, J., & Riquelme, J. C. (2019). A framework for evaluating land use and land cover classification using convolutional neural networks. *Remote Sensing*, 11(3), 274.
- Chadid, M. A., Dávalos, L. M., Molina, J., & Armenteras, D. (2015). A Bayesian spatial model highlights distinct dynamics in deforestation from coca and pastures in an Andean biodiversity hotspot. *Forests*, 6(11), 3828–3846.
- Chan, J. C.-W., Chan, K.-P., & Yeh, A. G.-O. (2001). Detecting the nature of change in an urban environment: A comparison of machine learning algorithms. *Photogrammetric Engineering and Remote Sensing*, 67(2), 213–226.
- Crooks, A. T. (2006). *Exploring cities using agent-based models and GIS*.

- Freund, Y., & Schapire, R. E. (1996). Experiments with a new boosting algorithm. *Icml*, 96, 148–156.
- Guo, Q., Kelly, M., Gong, P., & Liu, D. (2007). An Object-Based Classification Approach in Mapping Tree Mortality Using High Spatial Resolution Imagery. *GIScience & Remote Sensing*, 44(1), 24–47. <https://doi.org/10.2747/1548-1603.44.1.24>
- Hsu, C.-W., Chang, C.-C., & Lin, C.-J. (2003). *A practical guide to support vector classification*. Taipei, Taiwan.
- Islam, M. (2020). *A STUDY ON IMPACTS, CONSTRUCTION CHALLENGES AND OVERCOMES OF PADMA MULTIPURPOSE BRIDGE, BANGLADESH*. 8.
- Jensen, J. R. (1981). Urban Change Detection Mapping Using Landsat Digital Data. *The American Cartographer*, 8(2), 127–147. <https://doi.org/10.1559/152304081784447318>
- Kamusoko, C. (2019). *Remote Sensing Image Classification in R*. Springer Singapore. <https://doi.org/10.1007/978-981-13-8012-9>
- Kavzoglu, T., & Mather, P. M. (2003). The use of backpropagating artificial neural networks in land cover classification. *International Journal of Remote Sensing*, 24(23), 4907–4938.
- Kim, H., & Koehler, G. J. (1994). An investigation on the conditions of pruning an induced decision tree. *European Journal of Operational Research*, 77(1), 82–95.
- Kuhn, M., & Johnson, K. (2013). *Applied Predictive Modeling*. Springer New York. <https://doi.org/10.1007/978-1-4614-6849-3>
- Li, C., Wang, J., Wang, L., Hu, L., & Gong, P. (2014). Comparison of Classification Algorithms and Training Sample Sizes in Urban Land Classification with Landsat Thematic Mapper Imagery. *Remote Sensing*, 6(2), 964–983. <https://doi.org/10.3390/rs6020964>

- Lu, D., Mausel, P., Brondízio, E., & Moran, E. (2004). Change detection techniques. *International Journal of Remote Sensing*, 25(12), 2365–2401. <https://doi.org/10.1080/0143116031000139863>
- Mandelbrot, B. B., & Mandelbrot, B. B. (1982). *The fractal geometry of nature* (Vol. 1). WH freeman New York.
- Miao, X., Heaton, J. S., Zheng, S., Charlet, D. A., & Liu, H. (2012). Applying tree-based ensemble algorithms to the classification of ecological zones using multi-temporal multi-source remote-sensing data. *International Journal of Remote Sensing*, 33(6), 1823–1849.
- Osei-Bryson, K.-M. (2007). Post-pruning in decision tree induction using multiple performance measures. *Computers & Operations Research*, 34(11), 3331–3345.
- Pontius Jr, R. G., & Schneider, L. C. (2001). Land-cover change model validation by an ROC method for the Ipswich watershed, Massachusetts, USA. *Agriculture, Ecosystems & Environment*, 85(1–3), 239–248.
- Rahman, A., & Khondker, B. H. (2022). *Economic Cost-Benefit Analysis: Padma Bridge Project*. 38.
- Raihan, S., & Khondker, B. H. (n.d.). *Estimating the Economic Impacts of the Padma Bridge in Bangladesh*. 66.
- Rodriguez-Galiano, V. F., Chica-Olmo, M., Abarca-Hernandez, F., Atkinson, P. M., & Jeganathan, C. (2012). Random Forest classification of Mediterranean land cover using multi-seasonal imagery and multi-seasonal texture. *Remote Sensing of Environment*, 121, 93–107.
- Schneider, A., Friedl, M. A., McIver, D. K., & Woodcock, C. E. (2003). Mapping urban areas by fusing multiple sources of coarse resolution remotely sensed data. *Photogrammetric Engineering & Remote Sensing*, 69(12), 1377–1386.
- Sesnie, S. E., Gessler, P. E., Finegan, B., & Thessler, S. (2008). Integrating Landsat TM and SRTM-DEM derived variables with decision trees for habitat classification

- and change detection in complex neotropical environments. *Remote Sensing of Environment*, 112(5), 2145–2159.
- Sham, S. H. R., & Tapley, M. J. (n.d.). *The design of Padma Multipurpose Bridge – challenges and solutions in design of the river spans*. 10.
- Shen, G. (2002). Fractal dimension and fractal growth of urbanized areas. *International Journal of Geographical Information Science*, 16(5), 419–437.
- Soares-Filho, B. S., Coutinho Cerqueira, G., & Lopes Pennachin, C. (2002). dinamica— A stochastic cellular automata model designed to simulate the landscape dynamics in an Amazonian colonization frontier. *Ecological Modelling*, 154(3), 217–235. [https://doi.org/10.1016/S0304-3800\(02\)00059-5](https://doi.org/10.1016/S0304-3800(02)00059-5)
- Tso, B., & Mather, P. M. (2009). *Classification methods for remotely sensed data* (2nd ed). CRC Press.
- Wu, Q., Li, H., Wang, R., Paulussen, J., He, Y., Wang, M., Wang, B., & Wang, Z. (2006). Monitoring and predicting land use change in Beijing using remote sensing and GIS. *Landscape and Urban Planning*, 78(4), 322–333. <https://doi.org/10.1016/j.landurbplan.2005.10.002>
- Zaman, M., Shibuya, M., Okuno, K., & Horita, M. (2006). The Padma Bridge Feasibility Study: Social/Resettlement Impact Assessments. 建設マネジメント研究論文集 , 13, 51–61. <https://doi.org/10.2208/procm.13.51>
- Zhang, R., & Ma, J. (2008). An improved SVM method P-SVM for classification of remotely sensed data. *International Journal of Remote Sensing*, 29(20), 6029–6036.

Appendix

Metadata of the satellite image

The metadata for the Landsat09 image of 2022 from Padma region:

```
GROUP = LANDSAT_METADATA_FILE
GROUP = PRODUCT_CONTENTS
  ORIGIN = "Image courtesy of the U.S. Geological Survey"
  DIGITAL_OBJECT_IDENTIFIER = "https://doi.org/10.5066/P9OGBGM6"
  LANDSAT_PRODUCT_ID = "LC09_L2SP_138043_20220319_20220323_02_T1"
  PROCESSING_LEVEL = "L2SP"
  COLLECTION_NUMBER = 02
  COLLECTION_CATEGORY = "T1"
  OUTPUT_FORMAT = "GEOTIFF"
  FILE_NAME_BAND_1 = "LC09_L2SP_138043_20220319_20220323_02_T1_SR_B1.TIF"
  FILE_NAME_BAND_2 = "LC09_L2SP_138043_20220319_20220323_02_T1_SR_B2.TIF"
  FILE_NAME_BAND_3 = "LC09_L2SP_138043_20220319_20220323_02_T1_SR_B3.TIF"
  FILE_NAME_BAND_4 = "LC09_L2SP_138043_20220319_20220323_02_T1_SR_B4.TIF"
  FILE_NAME_BAND_5 = "LC09_L2SP_138043_20220319_20220323_02_T1_SR_B5.TIF"
  FILE_NAME_BAND_6 = "LC09_L2SP_138043_20220319_20220323_02_T1_SR_B6.TIF"
  FILE_NAME_BAND_7 = "LC09_L2SP_138043_20220319_20220323_02_T1_SR_B7.TIF"
  FILE_NAME_BAND_ST_B10 = "LC09_L2SP_138043_20220319_20220323_02_T1_ST_B10.TIF"
  FILE_NAME_THERMAL_RADIANCE = "LC09_L2SP_138043_20220319_20220323_02_T1_ST_TRAD.TIF"
  FILE_NAME_UPWELL_RADIANCE = "LC09_L2SP_138043_20220319_20220323_02_T1_ST_URAD.TIF"
  FILE_NAME_DOWNWELL_RADIANCE = "LC09_L2SP_138043_20220319_20220323_02_T1_ST_DRAD.TIF"
  FILE_NAME_ATMOSPHERIC_TRANSMITTANCE = "LC09_L2SP_138043_20220319_20220323_02_T1_ST_ATRAN.TIF"
  FILE_NAME_EMISSIVITY = "LC09_L2SP_138043_20220319_20220323_02_T1_ST_EMIS.TIF"
  FILE_NAME_EMISSIVITY_STDEV = "LC09_L2SP_138043_20220319_20220323_02_T1_ST_EMSD.TIF"
  FILE_NAME_CLOUD_DISTANCE = "LC09_L2SP_138043_20220319_20220323_02_T1_ST_CDIST.TIF"
  FILE_NAME_QUALITY_L2_AEROSOL = "LC09_L2SP_138043_20220319_20220323_02_T1_SR_QA_AEROSOL.TIF"
  FILE_NAME_QUALITY_L2_SURFACE_TEMPERATURE = "LC09_L2SP_138043_20220319_20220323_02_T1_ST_QA.TIF"
  FILE_NAME_QUALITY_L1_PIXEL = "LC09_L2SP_138043_20220319_20220323_02_T1_QA_PIXEL.TIF"
  FILE_NAME_QUALITY_L1_RADIOMETRIC SATURATION = "LC09_L2SP_138043_20220319_20220323_02_T1_QA_RADSAT.TIF"
  FILE_NAME_ANGLE_COEFFICIENT = "LC09_L2SP_138043_20220319_20220323_02_T1_ANG.txt"
  FILE_NAME_METADATA_ODL = "LC09_L2SP_138043_20220319_20220323_02_T1_MTL.txt"
  FILE_NAME_METADATA_XML = "LC09_L2SP_138043_20220319_20220323_02_T1_MTL.xml"
  DATA_TYPE_BAND_1 = "UINT16"
  DATA_TYPE_BAND_2 = "UINT16"
  DATA_TYPE_BAND_3 = "UINT16"
  DATA_TYPE_BAND_4 = "UINT16"
  DATA_TYPE_BAND_5 = "UINT16"
  DATA_TYPE_BAND_6 = "UINT16"
  DATA_TYPE_BAND_7 = "UINT16"
  DATA_TYPE_BAND_ST_B10 = "UINT16"
  DATA_TYPE_THERMAL_RADIANCE = "INT16"
```

```

DATA_TYPE_UPWELL_RADIANCE = "INT16"
DATA_TYPE_DOWNWELL_RADIANCE = "INT16"
DATA_TYPE_ATMOSPHERIC_TRANSMITTANCE = "INT16"
DATA_TYPE_EMISSIVITY = "INT16"
DATA_TYPE_EMISSIVITY_STDEV = "INT16"
DATA_TYPE_CLOUD_DISTANCE = "INT16"
DATA_TYPE_QUALITY_L2_AEROSOL = "UINT8"
DATA_TYPE_QUALITY_L2_SURFACE_TEMPERATURE = "INT16"
DATA_TYPE_QUALITY_L1_PIXEL = "UINT16"
DATA_TYPE_QUALITY_L1_RADIOMETRIC_SATURATION = "UINT16"
END_GROUP = PRODUCT_CONTENTS
GROUP = IMAGE_ATTRIBUTES
SPACECRAFT_ID = "LANDSAT_9"
SENSOR_ID = "OLI_TIRS"
WRS_TYPE = 2
WRS_PATH = 138
WRS_ROW = 43
NADIR_OFFNADIR = "NADIR"
TARGET_WRS_PATH = 138
TARGET_WRS_ROW = 43
DATE_ACQUIRED = 2022-03-19
SCENE_CENTER_TIME = "04:30:36.3615950Z"
STATION_ID = "LGN"
CLOUD_COVER = 0.01
CLOUD_COVER_LAND = 0.01
IMAGE_QUALITY_OLI = 9
IMAGE_QUALITY_TIRS = 9
SATURATION_BAND_1 = "N"
SATURATION_BAND_2 = "N"
SATURATION_BAND_3 = "N"
SATURATION_BAND_4 = "N"
SATURATION_BAND_5 = "N"
SATURATION_BAND_6 = "N"
SATURATION_BAND_7 = "Y"
SATURATION_BAND_8 = "N"
SATURATION_BAND_9 = "N"
ROLL_ANGLE = -0.001
SUN_AZIMUTH = 132.23140113
SUN_ELEVATION = 55.01387940
EARTH_SUN_DISTANCE = 0.9954128
END_GROUP = IMAGE_ATTRIBUTES
GROUP = PROJECTION_ATTRIBUTES
MAP_PROJECTION = "UTM"
DATUM = "WGS84"
ELLIPSOID = "WGS84"
UTM_ZONE = 45
GRID_CELL_SIZE_REFLECTIVE = 30.00
GRID_CELL_SIZE_THERMAL = 30.00
REFLECTIVE_LINES = 7691
REFLECTIVE_SAMPLES = 7541
THERMAL_LINES = 7691
THERMAL_SAMPLES = 7541
ORIENTATION = "NORTH_UP"
CORNER_UL_LAT_PRODUCT = 25.60155
CORNER_UL_LON_PRODUCT = 88.07950
CORNER_UR_LAT_PRODUCT = 25.56761
CORNER_UR_LON_PRODUCT = 90.33012
CORNER_LL_LAT_PRODUCT = 23.51832
CORNER_LL_LON_PRODUCT = 88.06181
CORNER_LR_LAT_PRODUCT = 23.48748

```

```

CORNER_LR_LON_PRODUCT = 90.27564
CORNER_UL_PROJECTION_X_PRODUCT = 608400.000
CORNER_UL_PROJECTION_Y_PRODUCT = 2832000.000
CORNER_UR_PROJECTION_X_PRODUCT = 834600.000
CORNER_UR_PROJECTION_Y_PRODUCT = 2832000.000
CORNER_LL_PROJECTION_X_PRODUCT = 608400.000
CORNER_LL_PROJECTION_Y_PRODUCT = 2601300.000
CORNER_LR_PROJECTION_X_PRODUCT = 834600.000
CORNER_LR_PROJECTION_Y_PRODUCT = 2601300.000
END_GROUP = PROJECTION_ATTRIBUTES
GROUP = LEVEL2_PROCESSING_RECORD
  ORIGIN = "Image courtesy of the U.S. Geological Survey"
  DIGITAL_OBJECT_IDENTIFIER = "https://doi.org/10.5066/P9OGBGM6"
  REQUEST_ID = "P701bp4r0ugyc_00036"
  LANDSAT_PRODUCT_ID = "LC09_L2SP_138043_20220319_20220323_02_T1"
  PROCESSING_LEVEL = "L2SP"
  OUTPUT_FORMAT = "GEOTIFF"
  DATE_PRODUCT_GENERATED = 2022-03-23T13:14:12Z
  PROCESSING_SOFTWARE_VERSION = "LPGS_15.6.0"
  ALGORITHM_SOURCE_SURFACE_REFLECTANCE = "LaSRC_1.5.0"
  DATA_SOURCE_OZONE = "MODIS"
  DATA_SOURCE_PRESSURE = "Calculated"
  DATA_SOURCE_WATER_VAPOR = "MODIS"
  DATA_SOURCE_AIR_TEMPERATURE = "MODIS"
  ALGORITHM_SOURCE_SURFACE_TEMPERATURE = "st_1.3.0"
  DATA_SOURCE_REANALYSIS = "GEOS-5 FP-IT"
END_GROUP = LEVEL2_PROCESSING_RECORD
GROUP = LEVEL2_SURFACE_REFLECTANCE_PARAMETERS
  REFLECTANCE_MAXIMUM_BAND_1 = 1.602213
  REFLECTANCE_MINIMUM_BAND_1 = -0.199972
  REFLECTANCE_MAXIMUM_BAND_2 = 1.602213
  REFLECTANCE_MINIMUM_BAND_2 = -0.199972
  REFLECTANCE_MAXIMUM_BAND_3 = 1.602213
  REFLECTANCE_MINIMUM_BAND_3 = -0.199972
  REFLECTANCE_MAXIMUM_BAND_4 = 1.602213
  REFLECTANCE_MINIMUM_BAND_4 = -0.199972
  REFLECTANCE_MAXIMUM_BAND_5 = 1.602213
  REFLECTANCE_MINIMUM_BAND_5 = -0.199972
  REFLECTANCE_MAXIMUM_BAND_6 = 1.602213
  REFLECTANCE_MINIMUM_BAND_6 = -0.199972
  REFLECTANCE_MAXIMUM_BAND_7 = 1.602213
  REFLECTANCE_MINIMUM_BAND_7 = -0.199972
  QUANTIZE_CAL_MAX_BAND_1 = 65535
  QUANTIZE_CAL_MIN_BAND_1 = 1
  QUANTIZE_CAL_MAX_BAND_2 = 65535
  QUANTIZE_CAL_MIN_BAND_2 = 1
  QUANTIZE_CAL_MAX_BAND_3 = 65535
  QUANTIZE_CAL_MIN_BAND_3 = 1
  QUANTIZE_CAL_MAX_BAND_4 = 65535
  QUANTIZE_CAL_MIN_BAND_4 = 1
  QUANTIZE_CAL_MAX_BAND_5 = 65535
  QUANTIZE_CAL_MIN_BAND_5 = 1
  QUANTIZE_CAL_MAX_BAND_6 = 65535
  QUANTIZE_CAL_MIN_BAND_6 = 1
  QUANTIZE_CAL_MAX_BAND_7 = 65535
  QUANTIZE_CAL_MIN_BAND_7 = 1
  REFLECTANCE_MULT_BAND_1 = 2.75e-05
  REFLECTANCE_MULT_BAND_2 = 2.75e-05
  REFLECTANCE_MULT_BAND_3 = 2.75e-05
  REFLECTANCE_MULT_BAND_4 = 2.75e-05

```

```

REFLECTANCE_MULT_BAND_5 = 2.75e-05
REFLECTANCE_MULT_BAND_6 = 2.75e-05
REFLECTANCE_MULT_BAND_7 = 2.75e-05
REFLECTANCE_ADD_BAND_1 = -0.2
REFLECTANCE_ADD_BAND_2 = -0.2
REFLECTANCE_ADD_BAND_3 = -0.2
REFLECTANCE_ADD_BAND_4 = -0.2
REFLECTANCE_ADD_BAND_5 = -0.2
REFLECTANCE_ADD_BAND_6 = -0.2
REFLECTANCE_ADD_BAND_7 = -0.2
END_GROUP = LEVEL2_SURFACE_REFLECTANCE_PARAMETERS
GROUP = LEVEL2_SURFACE_TEMPERATURE_PARAMETERS
TEMPERATURE_MAXIMUM_BAND_ST_B10 = 372.999941
TEMPERATURE_MINIMUM_BAND_ST_B10 = 149.003418
QUANTIZE_CAL_MAXIMUM_BAND_ST_B10 = 65535
QUANTIZE_CAL_MINIMUM_BAND_ST_B10 = 1
TEMPERATURE_MULT_BAND_ST_B10 = 0.00341802
TEMPERATURE_ADD_BAND_ST_B10 = 149.0
END_GROUP = LEVEL2_SURFACE_TEMPERATURE_PARAMETERS
GROUP = LEVEL1_PROCESSING_RECORD
ORIGIN = "Image courtesy of the U.S. Geological Survey"
DIGITAL_OBJECT_IDENTIFIER = "https://doi.org/10.5066/P975CC9B"
REQUEST_ID = "P501bp4pokdxw_00036"
LANDSAT_SCENE_ID = "LC091380432022078LGN00"
LANDSAT_PRODUCT_ID = "LC09_L1TP_138043_20220319_20220322_02_T1"
PROCESSING_LEVEL = "L1TP"
COLLECTION_CATEGORY = "T1"
OUTPUT_FORMAT = "GEOTIFF"
DATE_PRODUCT_GENERATED = 2022-03-22T20:20:58Z
PROCESSING_SOFTWARE_VERSION = "LPGS_15.6.0"
FILE_NAME_BAND_1 = "LC09_L1TP_138043_20220319_20220322_02_T1_B1.TIF"
FILE_NAME_BAND_2 = "LC09_L1TP_138043_20220319_20220322_02_T1_B2.TIF"
FILE_NAME_BAND_3 = "LC09_L1TP_138043_20220319_20220322_02_T1_B3.TIF"
FILE_NAME_BAND_4 = "LC09_L1TP_138043_20220319_20220322_02_T1_B4.TIF"
FILE_NAME_BAND_5 = "LC09_L1TP_138043_20220319_20220322_02_T1_B5.TIF"
FILE_NAME_BAND_6 = "LC09_L1TP_138043_20220319_20220322_02_T1_B6.TIF"
FILE_NAME_BAND_7 = "LC09_L1TP_138043_20220319_20220322_02_T1_B7.TIF"
FILE_NAME_BAND_8 = "LC09_L1TP_138043_20220319_20220322_02_T1_B8.TIF"
FILE_NAME_BAND_9 = "LC09_L1TP_138043_20220319_20220322_02_T1_B9.TIF"
FILE_NAME_BAND_10 = "LC09_L1TP_138043_20220319_20220322_02_T1_B10.TIF"
FILE_NAME_BAND_11 = "LC09_L1TP_138043_20220319_20220322_02_T1_B11.TIF"
FILE_NAME_QUALITY_L1_PIXEL = "LC09_L1TP_138043_20220319_20220322_02_T1_QA_PIXEL.TIF" =
FILE_NAME_QUALITY_L1_RADIOMETRIC SATURATION =
"LC09_L1TP_138043_20220319_20220322_02_T1_QA_RADSAT.TIF" =
FILE_NAME_ANGLE_COEFFICIENT =
"LC09_L1TP_138043_20220319_20220322_02_T1_ANG.txt" =
FILE_NAME_ANGLE_SENSOR_AZIMUTH_BAND_4 =
"LC09_L1TP_138043_20220319_20220322_02_T1_VAA.TIF" =
FILE_NAME_ANGLE_SENSOR_ZENITH_BAND_4 =
"LC09_L1TP_138043_20220319_20220322_02_T1_VZA.TIF" =
FILE_NAME_ANGLE_SOLAR_AZIMUTH_BAND_4 =
"LC09_L1TP_138043_20220319_20220322_02_T1_SAA.TIF" =
FILE_NAME_ANGLE_SOLAR_ZENITH_BAND_4 =
"LC09_L1TP_138043_20220319_20220322_02_T1_SZA.TIF" =
FILE_NAME_METADATA_ODL =
"LC09_L1TP_138043_20220319_20220322_02_T1_MTL.txt" =
FILE_NAME_METADATA_XML =
"LC09_L1TP_138043_20220319_20220322_02_T1_MTL.xml" =
FILE_NAME_CPF = "LC09CPF_20220319_20220331_02.05"

```

```

FILE_NAME_BPF_OLI = "LO9BPF20220319040714_20220319055256.01"
FILE_NAME_BPF_TIRS = "LT9BPF20220319040223_20220319045807.01"
FILE_NAME_RLUT = "LC09RLUT_20220319_20531231_02_08.h5"
DATA_SOURCE_ELEVATION = "GLS2000"
GROUND_CONTROL_POINTS_VERSION = 5
GROUND_CONTROL_POINTS_MODEL = 398
GEOMETRIC_RMSE_MODEL = 5.978
GEOMETRIC_RMSE_MODEL_Y = 4.345
GEOMETRIC_RMSE_MODEL_X = 4.106
GROUND_CONTROL_POINTS_VERIFY = 68
GEOMETRIC_RMSE_VERIFY = 1.728
END_GROUP = LEVEL1_PROCESSING_RECORD
GROUP = LEVEL1_MIN_MAX_RADIANCE
RADIANCE_MAXIMUM_BAND_1 = 766.11493
RADIANCE_MINIMUM_BAND_1 = -63.26602
RADIANCE_MAXIMUM_BAND_2 = 786.86060
RADIANCE_MINIMUM_BAND_2 = -64.97920
RADIANCE_MAXIMUM_BAND_3 = 723.02130
RADIANCE_MINIMUM_BAND_3 = -59.70734
RADIANCE_MAXIMUM_BAND_4 = 612.81738
RADIANCE_MINIMUM_BAND_4 = -50.60666
RADIANCE_MAXIMUM_BAND_5 = 375.97250
RADIANCE_MINIMUM_BAND_5 = -31.04793
RADIANCE_MAXIMUM_BAND_6 = 93.92427
RADIANCE_MINIMUM_BAND_6 = -7.75630
RADIANCE_MAXIMUM_BAND_7 = 31.71428
RADIANCE_MINIMUM_BAND_7 = -2.61898
RADIANCE_MAXIMUM_BAND_8 = 693.68884
RADIANCE_MINIMUM_BAND_8 = -57.28505
RADIANCE_MAXIMUM_BAND_9 = 155.96777
RADIANCE_MINIMUM_BAND_9 = -12.87987
RADIANCE_MAXIMUM_BAND_10 = 25.00330
RADIANCE_MINIMUM_BAND_10 = 0.10038
RADIANCE_MAXIMUM_BAND_11 = 22.97172
RADIANCE_MINIMUM_BAND_11 = 0.10035
END_GROUP = LEVEL1_MIN_MAX_RADIANCE
GROUP = LEVEL1_MIN_MAX_REFLECTANCE
REFLECTANCE_MAXIMUM_BAND_1 = 1.210700
REFLECTANCE_MINIMUM_BAND_1 = -0.099980
REFLECTANCE_MAXIMUM_BAND_2 = 1.210700
REFLECTANCE_MINIMUM_BAND_2 = -0.099980
REFLECTANCE_MAXIMUM_BAND_3 = 1.210700
REFLECTANCE_MINIMUM_BAND_3 = -0.099980
REFLECTANCE_MAXIMUM_BAND_4 = 1.210700
REFLECTANCE_MINIMUM_BAND_4 = -0.099980
REFLECTANCE_MAXIMUM_BAND_5 = 1.210700
REFLECTANCE_MINIMUM_BAND_5 = -0.099980
REFLECTANCE_MAXIMUM_BAND_6 = 1.210700
REFLECTANCE_MINIMUM_BAND_6 = -0.099980
REFLECTANCE_MAXIMUM_BAND_7 = 1.210700
REFLECTANCE_MINIMUM_BAND_7 = -0.099980
REFLECTANCE_MAXIMUM_BAND_8 = 1.210700
REFLECTANCE_MINIMUM_BAND_8 = -0.099980
REFLECTANCE_MAXIMUM_BAND_9 = 1.210700
REFLECTANCE_MINIMUM_BAND_9 = -0.099980
END_GROUP = LEVEL1_MIN_MAX_REFLECTANCE
GROUP = LEVEL1_MIN_MAX_PIXEL_VALUE
QUANTIZE_CAL_MAX_BAND_1 = 65535
QUANTIZE_CAL_MIN_BAND_1 = 1
QUANTIZE_CAL_MAX_BAND_2 = 65535

```

```

QUANTIZE_CAL_MIN_BAND_2 = 1
QUANTIZE_CAL_MAX_BAND_3 = 65535
QUANTIZE_CAL_MIN_BAND_3 = 1
QUANTIZE_CAL_MAX_BAND_4 = 65535
QUANTIZE_CAL_MIN_BAND_4 = 1
QUANTIZE_CAL_MAX_BAND_5 = 65535
QUANTIZE_CAL_MIN_BAND_5 = 1
QUANTIZE_CAL_MAX_BAND_6 = 65535
QUANTIZE_CAL_MIN_BAND_6 = 1
QUANTIZE_CAL_MAX_BAND_7 = 65535
QUANTIZE_CAL_MIN_BAND_7 = 1
QUANTIZE_CAL_MAX_BAND_8 = 65535
QUANTIZE_CAL_MIN_BAND_8 = 1
QUANTIZE_CAL_MAX_BAND_9 = 65535
QUANTIZE_CAL_MIN_BAND_9 = 1
QUANTIZE_CAL_MAX_BAND_10 = 65535
QUANTIZE_CAL_MIN_BAND_10 = 1
QUANTIZE_CAL_MAX_BAND_11 = 65535
QUANTIZE_CAL_MIN_BAND_11 = 1
END_GROUP = LEVEL1_MIN_MAX_PIXEL_VALUE
GROUP = LEVEL1_RADIOMETRIC_RESCALING
RADIANCE_MULT_BAND_1 = 1.2656E-02
RADIANCE_MULT_BAND_2 = 1.2998E-02
RADIANCE_MULT_BAND_3 = 1.1944E-02
RADIANCE_MULT_BAND_4 = 1.0123E-02
RADIANCE_MULT_BAND_5 = 6.2108E-03
RADIANCE_MULT_BAND_6 = 1.5516E-03
RADIANCE_MULT_BAND_7 = 5.2390E-04
RADIANCE_MULT_BAND_8 = 1.1459E-02
RADIANCE_MULT_BAND_9 = 2.5765E-03
RADIANCE_MULT_BAND_10 = 3.8000E-04
RADIANCE_MULT_BAND_11 = 3.4900E-04
RADIANCE_ADD_BAND_1 = -63.27867
RADIANCE_ADD_BAND_2 = -64.99220
RADIANCE_ADD_BAND_3 = -59.71928
RADIANCE_ADD_BAND_4 = -50.61678
RADIANCE_ADD_BAND_5 = -31.05414
RADIANCE_ADD_BAND_6 = -7.75785
RADIANCE_ADD_BAND_7 = -2.61950
RADIANCE_ADD_BAND_8 = -57.29651
RADIANCE_ADD_BAND_9 = -12.88245
RADIANCE_ADD_BAND_10 = 0.10000
RADIANCE_ADD_BAND_11 = 0.10000
REFLECTANCE_MULT_BAND_1 = 2.0000E-05
REFLECTANCE_MULT_BAND_2 = 2.0000E-05
REFLECTANCE_MULT_BAND_3 = 2.0000E-05
REFLECTANCE_MULT_BAND_4 = 2.0000E-05
REFLECTANCE_MULT_BAND_5 = 2.0000E-05
REFLECTANCE_MULT_BAND_6 = 2.0000E-05
REFLECTANCE_MULT_BAND_7 = 2.0000E-05
REFLECTANCE_MULT_BAND_8 = 2.0000E-05
REFLECTANCE_MULT_BAND_9 = 2.0000E-05
REFLECTANCE_ADD_BAND_1 = -0.100000
REFLECTANCE_ADD_BAND_2 = -0.100000
REFLECTANCE_ADD_BAND_3 = -0.100000
REFLECTANCE_ADD_BAND_4 = -0.100000
REFLECTANCE_ADD_BAND_5 = -0.100000
REFLECTANCE_ADD_BAND_6 = -0.100000
REFLECTANCE_ADD_BAND_7 = -0.100000
REFLECTANCE_ADD_BAND_8 = -0.100000

```

```

    REFLECTANCE_ADD_BAND_9 = -0.100000
END_GROUP = LEVEL1_RADIOMETRIC_RESCALING
GROUP = LEVEL1_THERMAL_CONSTANTS
    K1_CONSTANT_BAND_10 = 799.0284
    K2_CONSTANT_BAND_10 = 1329.2405
    K1_CONSTANT_BAND_11 = 475.6581
    K2_CONSTANT_BAND_11 = 1198.3494
END_GROUP = LEVEL1_THERMAL_CONSTANTS
GROUP = LEVEL1_PROJECTION_PARAMETERS
    MAP_PROJECTION = "UTM"
    DATUM = "WGS84"
    ELLIPSOID = "WGS84"
    UTM_ZONE = 45
    GRID_CELL_SIZE_PANCHROMATIC = 15.00
    GRID_CELL_SIZE_REFLECTIVE = 30.00
    GRID_CELL_SIZE_THERMAL = 30.00
    ORIENTATION = "NORTH_UP"
    RESAMPLING_OPTION = "CUBIC_CONVOLUTION"
END_GROUP = LEVEL1_PROJECTION_PARAMETERS
END_GROUP = LANDSAT_METADATA_FILE
END

```

R code for Classification

```

1  ##Calling Libraries##
2
3  library(sp)
4  library(rgdal)
5  library(raster)
6  library(reshape2)
7  library(grid)
8  library(gridExtra)
9  library(RStoolbox)
10 library(corrplot)
11 library(caret)
12 library(e1071)
13 library(rasterVis)
14 library(corrplot)
15 library(doParallel)
16
17 library(randomForest)
18
19
20 ##Setting Working Files##
21 setwd("F:\\ThesisAnalysis\\AnalysisV1\\AnalysisClassification1992\\Jamuna1992")
22 Jam92List <- list.files(getwd(),pattern="TIF$"),full.names = TRUE)
23 23
24 Jam92Stack <- stack(Jam92List)
25
26 Jam92Stack
27
28 png("Jamuna92TCI.png", width = 1920, height = 1920, units = "px")
29 Jamuna92 <- plotRGB(Jam92Stack, r=3, g=2, b=1, stretch="lin")
30 dev.off()
31

```

```

32  ##Loading training points##
33  taJam92 <- readOGR(getwd(), "TAJamuna92")
34  taJam92$Class <- as.factor(taJam92$Class)
35  str(taJam92)
36
37  ##Creating Spectral Profile##
38  Jam92Spec <- as.data.frame(extract(Jam92Stack,
39  taJam92))
40  head(Jam92Spec)
41  mrj92 <- aggregate(Jam92Spec, list(taJam92$Class), median,
42  na.rm=TRUE)
43  head(mrj92)
44  rownames(mrj92) <- mrj92[,1]
45  mrj92 <- mrj92[,-1]
46
47
48  ##Plotting Spectral Profile##
49  legColor <- c('green','yellow', 'red', 'cyan')
50  mrj92 <- as.matrix(mrj92)
51  svg("Jamuna92SP.svg")
52  plot(0, ylim=c(7700,23500), xlim=c(1,7), type='n', xlab="Band No",
53  ylab="Surface Reflectance")
54  for (i in 1:nrow(mrj92)){
55    lines(mrj92[i,], type="l", lwd=3, lty=1, col=legColor[i])}
56  legend("topleft", rownames(mrj92), cex = 0.8, col = legColor, lty =
57  1, lwd=3, bty="n")
58  dev.off()
59
60  ## Reflectance Dataframe ##
61  taJam92@data=data.frame(taJam92@data,Jam92Spec[match(rownames(taJam92
62  @data), rownames( Jam92Spec)),])
63  str(taJam92@data)
64
65  summary(taJam92@data)
66
67  ##Preparing Training Datasets##
68  seedValue <- 21
69  set.seed(seedValue)
70  inTraining <- createDataPartition(taJam92@data$Class, p=.80, list =
71  FALSE)
72  training <- taJam92@data[inTraining,]
73  testing <- taJam92@data[-inTraining,]
74  testing <- na.omit(taJam92@data[-inTraining,])
75  summary(training)
76  summary(testing)
77
78  ##Visualizing Training Quality##
79  skewnessValues <- apply(training[, 2:8], 2,
80  skewness)
81  skewnessValues
82  png("J92boxp.png", width = 1920,height =
83  1080, units= "px")
84  featurePlot(x=training[,2:8],y=training$Class
85  ,
86  plot="box",

```

```

80         scales=list(y=list(relation="free"),
81         x=list(rot=90)),
82         layout=c(3,3),
83         auto.key=list(columns=3))
84         dev.off()
86
87     bandCorrelations <- cor(training[, 2:8])
88     bandCorrelations
89
90     png("j92corr.png", width = 1920,height = 1080, units= "px")
91     corrplot(bandCorrelations, method = "circle", type = "lower",outline
92     = FALSE)
93     dev.off()
94
95     ##Model Tuning##
96     fitControl<- trainControl(method =
97     "repeatedcv",
98     number = 5,
99     repeats = 5)
100
101     StartTime <- proc.time()
102     rf_model<- train(Class~., data = training, method=
103     "rf",
104     trControl=fitControl, prox=TRUE,
105     fitbest =FALSE,
106     returnData=TRUE)
107     proc.time() - StartTime
108
109     print(rf_model)
110     plot(rf_model)
111     rf_model$finalModel
112     rf_varImp <- varImp(rf_model, compete = FALSE)
113
114     png("J92Varimp.png", width = 720,height =
115     600, units= "px") 113     plot(rf_varImp)
116     dev.off()
117
118     pred_rf <- predict(rf_model$finalModel, newdata = testing)
119     confusionMatrix(data = pred_rf, testing$Class)
120
121     Jamuna_rf_92 <-predict(Jam92Stack,rf_model)
122     writeRaster(Jamuna_rf_92, filename =
123     "Jamuna_rf_92_LULC.tif") 123
124     Jamuna_rf_92a <- gplot(Jamuna_rf_92) + geom_raster(aes(fill
125     =factor(value, labels=c(
126         "Agriculture", "Bareland", "Builtup", "Water")))) +
127     scale_fill_manual(values = c("green" , "yellow", "red","blue3"),name= "Land
128     Cover") + ggtitle("1992 Classification") +theme( plot.title =
129     element_text(lineheight=.4,face="bold")) + coord_equal()

```